



SP25: INNOVATION FOR IMPACT: 34591

DS-56-Advancing Operational Technology Training for Diplomatic Missions

Krishna Chaitanya, Vinoth Karunagaran, Ananmay Rajan, Thangamuthu Balaji

What is your understanding of the problem?

- Lack of “training” frameworks for Technical Security (TS) personnel on OT systems by OT security officers. (unable to be flexible in the integration of system types, operating systems and usability).
- Lack of understanding in applying the OT Audit and Compliance for all U.S missions overseas.

Who are the beneficiaries?

OT Security Officers	DoS IT Security Managers
Security Engineering Officers (SEOs)	Policy and Compliance Officers
Security Technical Specialists (STS)	Embassy Security Coordinators
Cybersecurity Engineers (OT Focus)	Cyber Threat Intelligence Analysts
Regional Security Officers (RSOs)	Training Content Developers
Facility Security Engineers (FSEs)	
Systems Integration Specialists	
OT Compliance Auditors	
Network Security Analysts (OT Division)	
Training Program Managers	
Diplomatic Security Technology Officers (DSTO)	
Information Systems Security Officers (ISSOs)	

What is the current "obvious" solution?

- Implementing an effective OT “Security Training Network System”.
- Build a comprehensive Framework for Technical Security (TS) personnel to learn about Operational Technology (OT) systems and different roles in OT and apply OT audit and compliance across all U.S missions overseas.
- Upgrading the OT systems to feature more modern security features but also adhere to existing cybersecurity policies and standards.
- Build a Training network that connects to the internet securely and also mimicking how all systems can connect in the same network.

The Mission Model Canvas

■ Key Partner

Phillip Miyasaka
(Problem Sponsor)

■ Key Activities

Build a Training Network
system

■ Beneficiaries

- Network Security Analysts
- OT Security Officers
- Security Technical Specialists
- Security Engineering Officers
- Cyber Threat Intelligence Analysts

■ Key Resources

- VMware (Virtual Machines)
- VPN Access (Openvpn or wireguard)
- Cloud Platforms (Azure, AWS, Google Cloud)
- Database (InfluxDB and MySQL)
- Partnership with federal agencies and private sector experts

■ Value Proposition

- Enable secure remote training for 14–28 personnel across U.S. global missions.
- Deploy virtual labs mirroring embassy OT networks with audit tools for DES, FISMA, and non-enterprise configurations.

■ Buy-in & Support

WIP

■ Deployment

WIP

■ Mission Budget

WIP

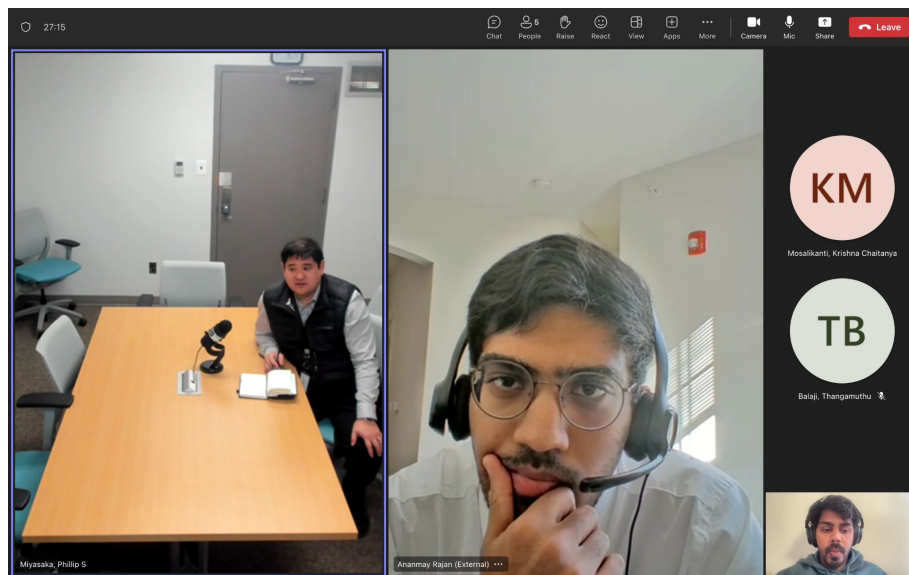
■ Achievement

WIP

Outreach Plan

Describe your plan for beneficiary discovery this week. Have you reached out to your project sponsor(s)? Is that meeting scheduled or has it happened already? How are you going to reach beneficiaries this week?

- Feb 6th - Had an Introduction Meeting with Phillip (DS-56)
- Potential Beneficiaries - To be interviewed next week starting with **10 beneficiaries**.



Team OT Guardians

Members

Ananmay Rajan, Vinoth Karunagaran, Thangamuthu Balaji, Krishna Chaitanya M

No. of beneficiaries Interviewed - 0 (still waiting)

Total no. of Interviews (Over all weeks) - 2 (With our problem Sponsor “Philip”)

Problem

- Requirement of a “platform” (Network system)
- Using standalone commercial methods of training (SANS, Udemy, OP Swat, CISA)
- More protection needed

Why it matters?

- There is no proper standard/method of training people about OT systems.
- Reliance on one computer (Can launch different things; commercial training, red vs blue teams, OT system simulations, How to identify vulnerabilities, etc)

Beneficiary Discovery

Hypotheses

What did you think beginning the week?

We began our week with an introductory meeting with our sponsor, gaining valuable insights and input on the requirements we need to work on.

Experiments

What did you do this week?

After the meeting, we had a brainstorming session on how to approach the requirements. We also created a draft for the Network Training System based on our understanding. Later, we prepared few interview questions for both the sponsor and the beneficiaries.

Results

What did you discover?

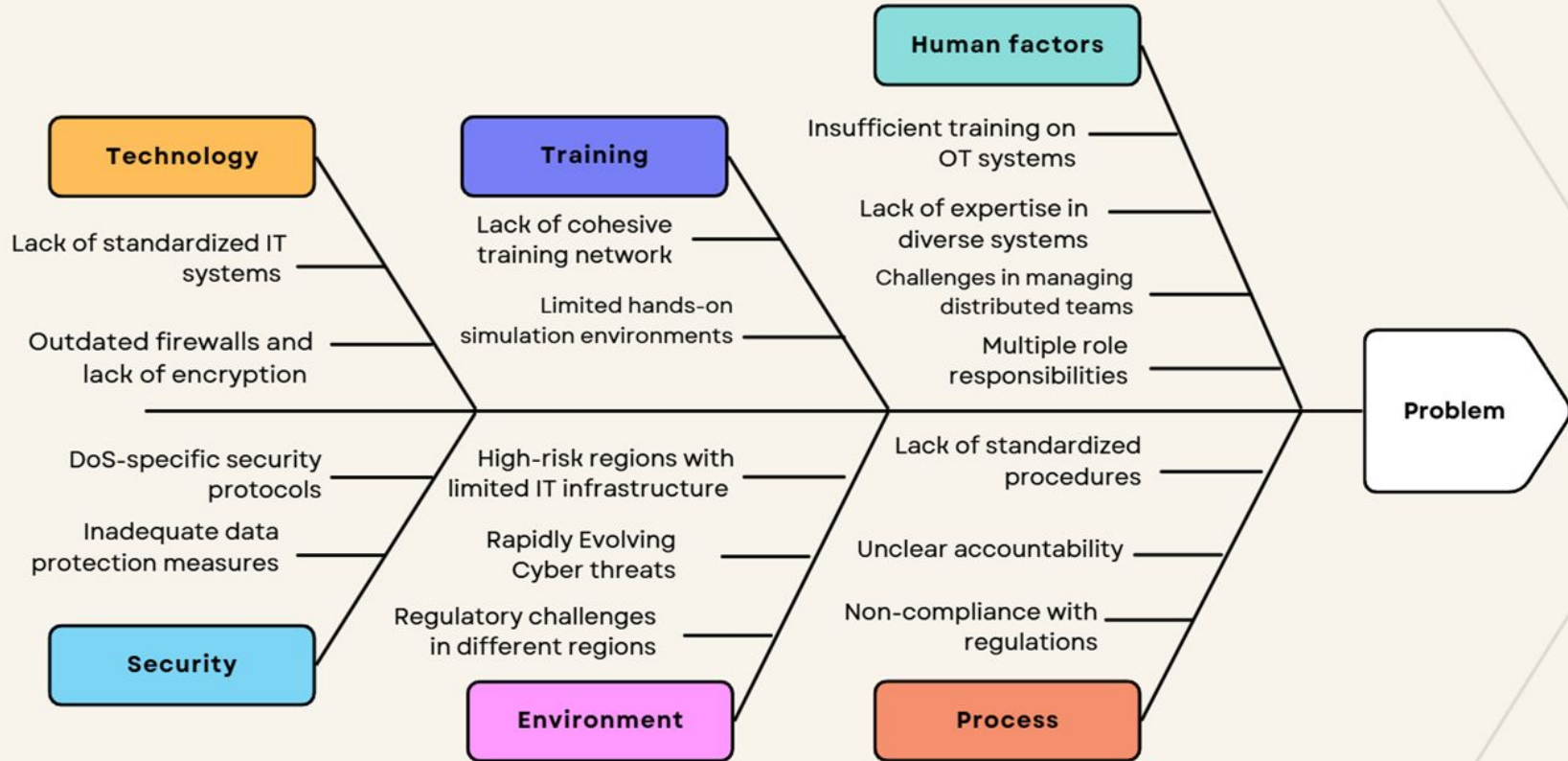
Focus on building a virtual OT network, adding security controls and testing for cyber threats. Our main beneficiaries include TS, physical security engineers, security auditors and DoS-related units unfamiliar with OT systems.

Actions

What will you do now? How will you proceed?

Our system needs to be “Robust”, “Scalable”, “Flexible”

Problem Diagram (Root Cause analysis for Operational Challenges)



Minimal Viable Product (MVP)

For this week (which meets minimum necessary requirements but is subject to change/improved in the future)

- A network system (Needs to be “Virtually Separated” - one for commercial use and the other for simulations)
- Ability to connect and cut off the “internet”, in case something bad happens
- Enough memory and storage for any simulations
- Basic I/O devices (Mouse, Keyboard)
- Basic security practices
- Connecting multiple systems simultaneously at the same time without any latency issues.
- Network system needs to be “robust” to support multiple operating systems.

What is it testing, data expected and data received. . .

Testing - A network system can be created with specific requirements

Data Expected - Previous training methods used

Data Received - Using standalone training methods

The Mission Model Canvas (updated)

■ Key Partner

Phillip Miyasaka
(Problem Sponsor)

■ Key Activities

Build a Training Network
system

■ Beneficiaries

- Technical Security Personnel
- Security Auditors
- People at the DoS (related units)
- IT teams
- Physical security engineers (Like our problem sponsor)

■ Key Resources

- VMware (Virtual Machines)
- VPN Access (Openvpn or wireguard)
- Cloud Platforms (Azure, AWS, Google Cloud)
- Database (InfluxDB and MySQL)
- Partnership with federal agencies and private sector experts

■ Value Proposition

- Tailored Training for OT related personnel across multiple systems simultaneously. (i.e. Lab environment)
- Simulated Real-World Scenarios.
- Remote Accessibility.

■ Buy-in & Support

WIP

■ Deployment

WIP

■ Mission Budget

WIP

■ Achievement

WIP

Plan for Next Week

What is your plan for next week?

Conduct Beneficiary interviews , Learn on building a network system.

Who will you talk to?

Regional cyber security Officers , Red and Blue Team, Supervisor of our problem sponsor

What will you do?

- Look for possible resources to create an Network System
- Conduct as many interviews as possible

What do you need to find out?

Speak with technical staff who handles network systems, labs and virtual machines and how it works at Indiana University, Bloomington

Week 4

Interview counts - 2

Chaoqi Zhang PhD student Computer Science @ IUB

Phillip (Our Sponsor)

Total no of interviews (so far) - 4

Updated Problem Statement

- Lack of training of OT systems for students/employees (Non - technical people)
- Hosting simulations of red vs blue team for OT systems (Technical People)



All under a single network system



Beneficiary Discovery (What we learnt from the beneficiary interviews)

1

Hypotheses

What did you think beginning the week?

We tried to “brainstorm” more ways on how can we build a “demo” of a network system with the free available tools and softwares.

2

Experiments

What did you do this week?

We had a meeting with Chaoqi Zhang (TA) to learn about virtual machines, operating systems and virtual imaging and various scripting tools needed to include various features in a network system.

3

Results

What did you discover?

We discovered that Linux is the best option to go forward when creating our demo. (Offers various customization options)

Docker is a valuable tool when it comes to scripting

4

Actions

What's next? How do you proceed?

Gather valuable inputs from our scheduled beneficiary interviews next week

Minimal Viable Product (MVP) (week 4)

For this week (which meets minimum necessary requirements but is subject to change/improved in the future)

- A network system (Needs to be “Virtually Separated” - one for commercial use and the other for simulations)
- Installing and learning Linux OS.
- Ability to connect and cut off the “internet”, in case something bad happens
- Enough memory and storage for any simulations
- Basic security practices
- Connecting multiple systems simultaneously at the same time without any latency issues.
- Network system needs to be “robust” to support multiple operating systems.

What is it testing, data expected and data received. . .

Testing - Can Linux based VM's support multiple users at once?

Data Expected - Yes, It can

Data Received - It can be possible if it meets a bunch of software/hardware requirements

The Mission Model Canvas (Week 4)

Key Partner

Phillip Miyasaka
(Problem Sponsor)

Key Activities

Build a Training Network
system

Value Proposition

- Tailored Training for OT related personnel across multiple systems simultaneously. (i.e. Lab environment)
- Simulated Real-World Scenarios.
- Remote Accessibility.
- Training for Non-Technical personnel (People who work at HVAC)

Buy-in & Support

WIP

Beneficiaries

- OT security personnel (Primary Beneficiary)
- Technical Security Personnel
- Security Auditors
- People at the DoS (related units)
- IT teams
- Physical security engineers (Like our problem sponsor)

Key Resources

- VMware (Virtual Machines)
- VPN Access (Openvpn or wireguard)
- Cloud Platforms (Azure, AWS, Google Cloud)
- Database (InfluxDB and MySQL)
- Partnership with federal agencies and private sector experts

Deployment

WIP

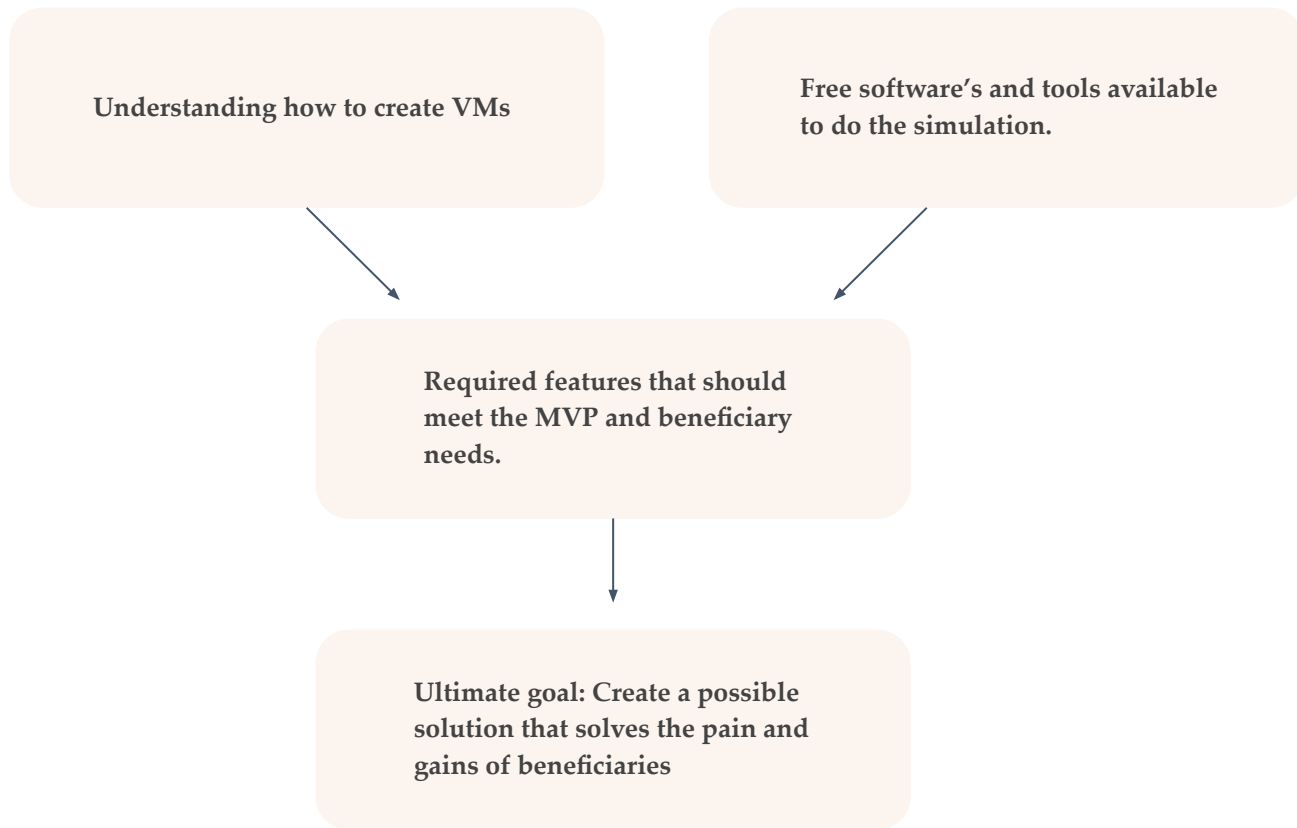
Mission Budget

WIP

Achievement

WIP

Choice Diagram



Plan for Next Week (Week 4)

What is your plan for next week?

Scheduled a meet with the Office Director of Technical Security Engineering (Tena Grest), possibly conduct more beneficiary interviews, have another meet with our sponsor, test out some of the free softwares

Who will you talk to?

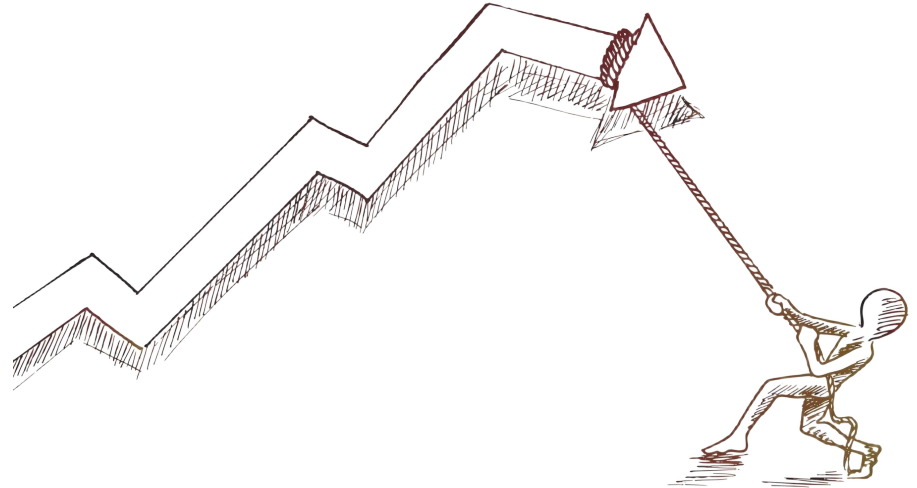
OT Security personnel, Regional cyber security Officers , Red and Blue Team in IT dept, Some of the employees who will undergo training

What will you do?

- Try to install and experiment with existing softwares/Tools for network systems using VirtualBox
- Conduct as many interviews as possible

What do you need to find out?

- Limitations and Advantages of using free softwares and Linux OS



Week 5 (March 1st)

Interviews this week: 3 (Tena Grest), (Brian)

Total Interview counts: 7



Updated Problem Statement

Engineers at the DoS and embassy staff worldwide lack standardized security training on OT systems.

Beneficiary Discovery Week 5 (What we learnt from the beneficiary interviews)

1

Hypotheses

What did you think beginning the week?

Does our idea have the necessary free softwares/tools to be implemented for a demo purpose?

2

Experiments

What did you do this week?

We had 2 interviews and learnt about the problem description and day to day activities of different units in the DoS

3

Results

What did you discover?

- The State Department lacks OT security expertise, seeks training for engineers
- Needs training environment/"sandbox" for vulnerability testing across distributed systems Faces challenges with limited in-house cybersecurity knowledge and centralized monitoring
- Specifically interested in red team/blue team exercises for attack simulation
- Critical need to build internal capacity for proactive OT security monitoring and threat mitigation

4

Actions

What's next? How do you proceed?

We plan to interview several people from the risk management unit in DoS to get more robust information and multiple data points.

The Mission Model Canvas (Week 5)

Key Partner

Phillip Miyasaka
(Problem Sponsor)
Tena Grest

Key Activities

Secondary Research on virtual network systems
Completed 7 rounds of interviews

Value Proposition

1. To audit OT systems across different US diplomatic missions
2. In-person security training (Springfield, VA)
3. Hands-on training to protect the organization's Network from cyber threats
4. Security Training for Non-Technical personnel

Buy-in & Support

WIP

Beneficiaries

1. OT security personnel (Primary Beneficiary)
2. People working at different US embassies (South Korea - Seoul, Thailand - Bangkok)
3. Technical Security Personnel
4. Different types of Engineers

Key Resources

- Red team/Blue Team Simulation Framework
- Establish basic network security testing environment.
- Internal Capacity Building Programs.

Deployment

WIP

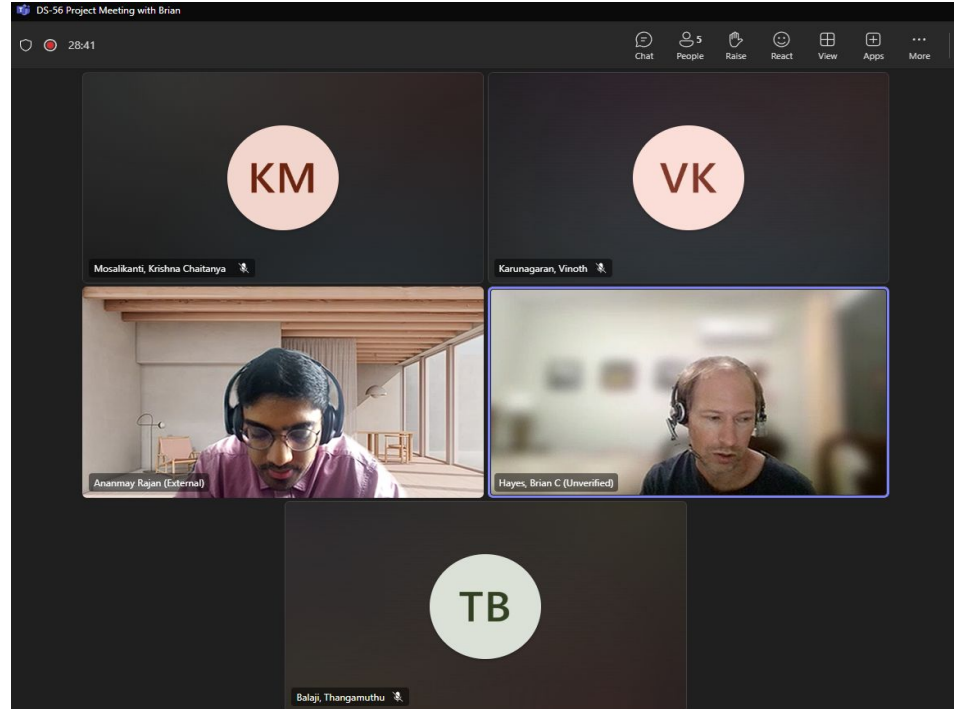
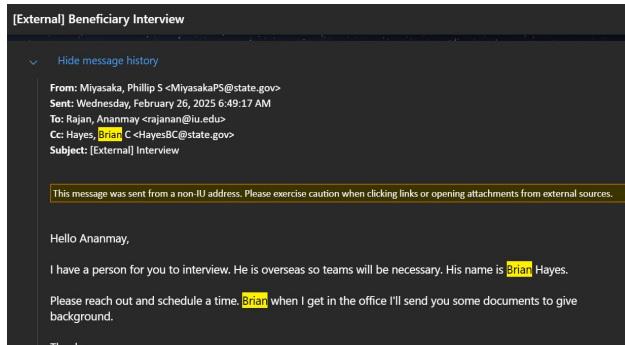
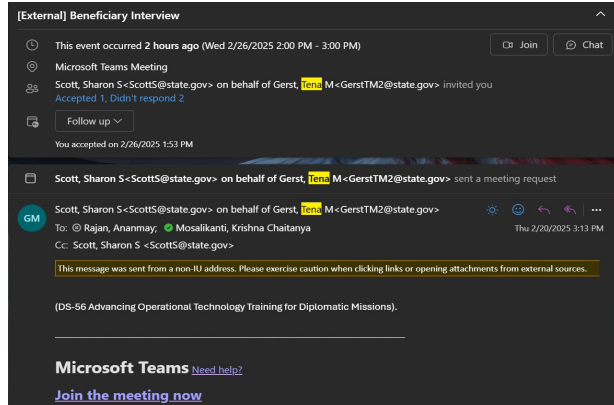
Mission Budget

They are open to investing in third party vendors

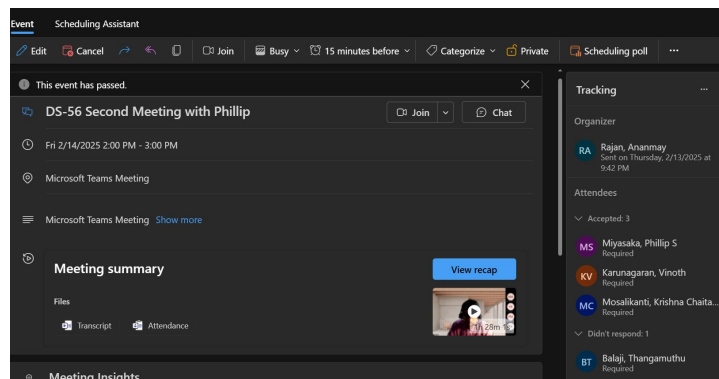
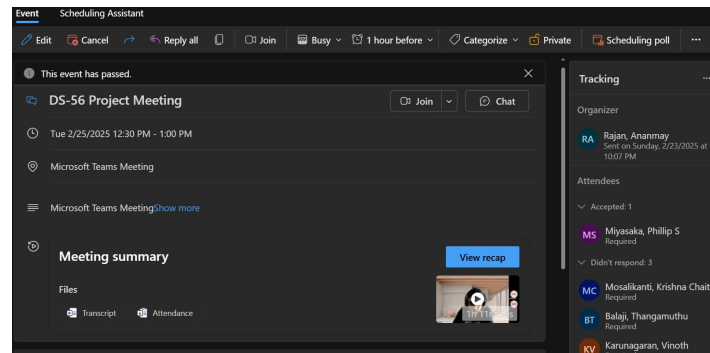
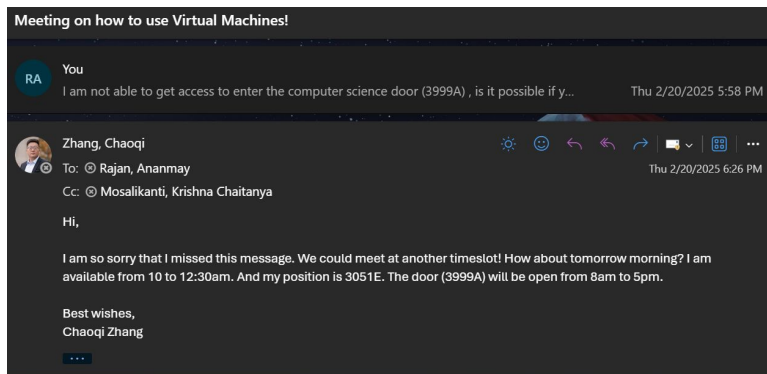
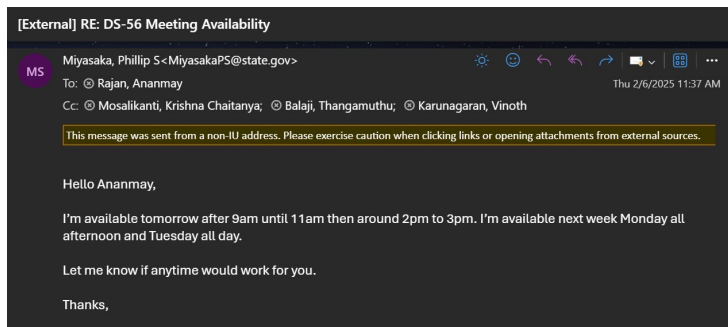
Achievement

WIP

Beneficiary Discovery Proof (Our discussion was virtual as they live in DC and Virginia)



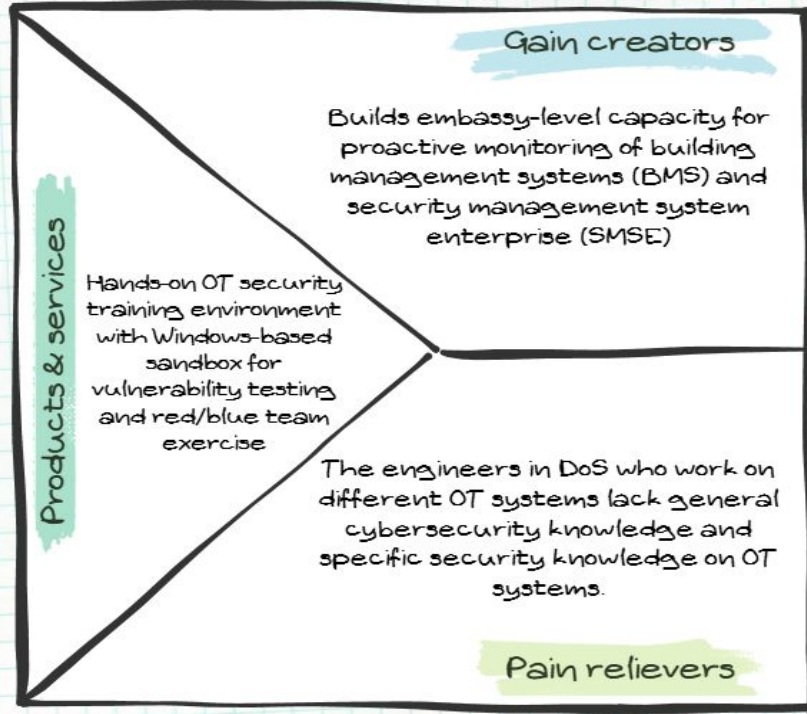
Beneficiary Discovery (Continued)



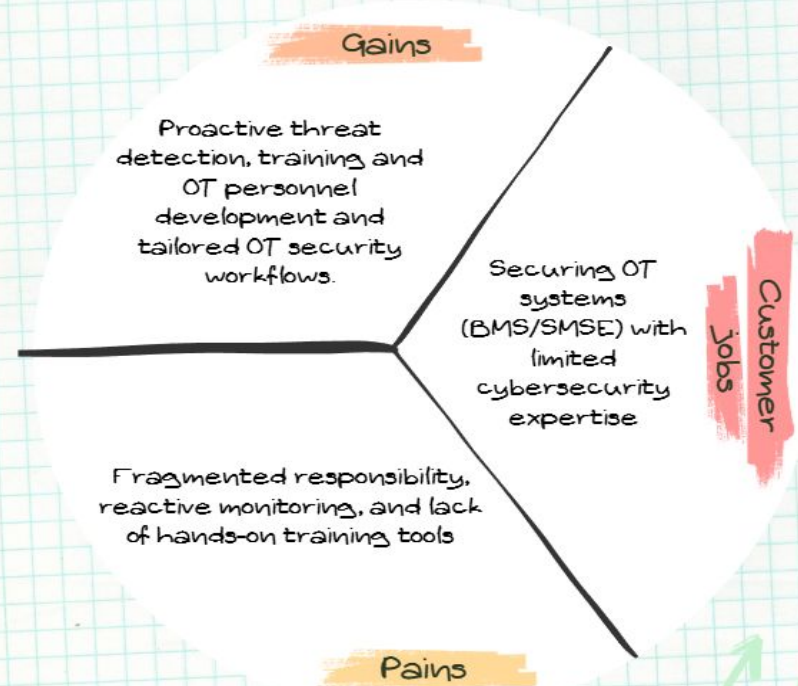
Value Proposition Canvases* + Beneficiary Archetypes*

Value Proposition Canvas

TENA



Value proposition



Customer segments

CL

Tena

Occupation: Operational Technology Security Officer

Location: Springfield, VA

Role: Operational Technology (OT) Security Trainer

ABOUT

Tena works at the State Department, helping engineers learn how to secure critical systems like building management and security networks. She enjoys solving tough problems in cybersecurity and mentoring young engineers while exploring new technologies.

GOALS

- Build a robust OT security training program for engineers at State Department posts worldwide.
- Develop a hands-on "sandbox" environment for testing vulnerabilities and simulating attacks/defenses.
- Ensure training aligns with real-world scenarios, such as red team/blue team exercises.

FRUSTRATION

- Lack of in-house cybersecurity knowledge specific to OT systems.
- Challenges in identifying the right tools and components for building a testing environment.

FAVORITE TOOLS & PLATFORMS

- Honeywell (HVAC systems)
- Genetec (camera monitoring)
- CISA Security Operations Center frameworks

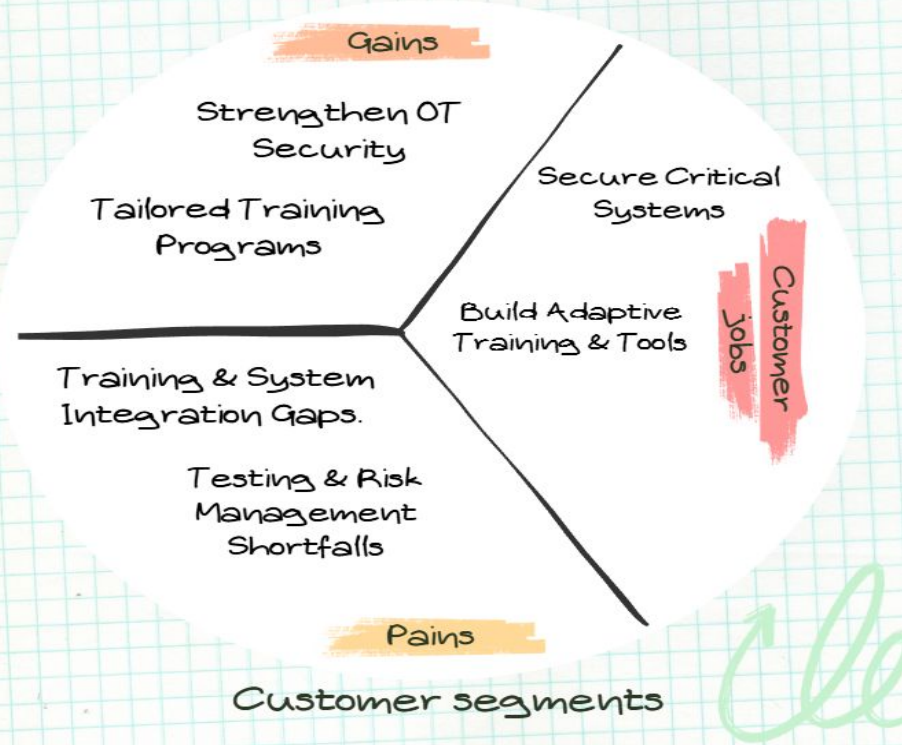
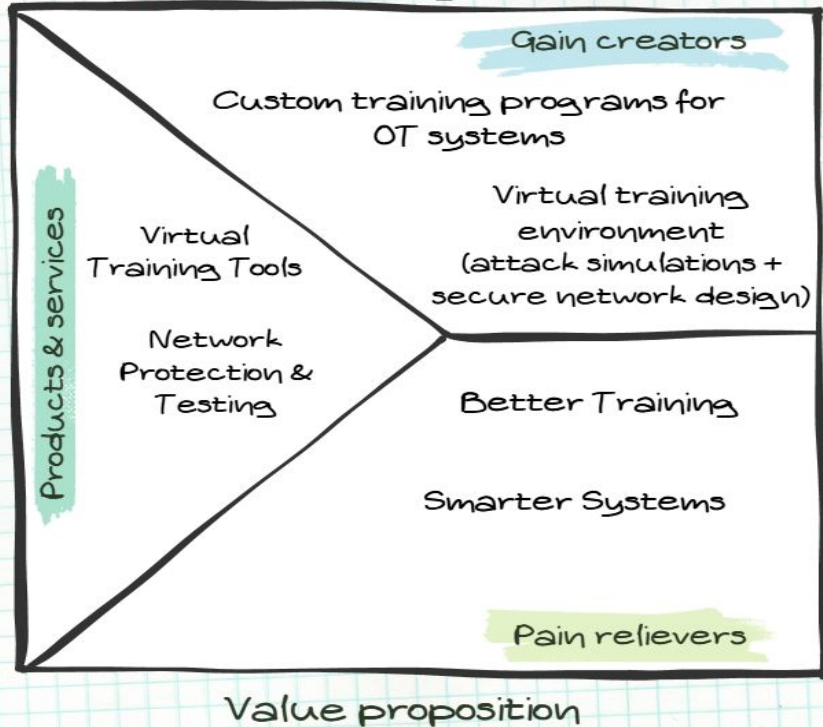
MOTIVATIONS

Tena wants to help engineers learn the skills they need to handle modern cybersecurity challenges. She believes giving them the right tools and training will make OT systems safer and stronger at State Department locations around the world.

Value Proposition Canvases* + Beneficiary Archetypes*

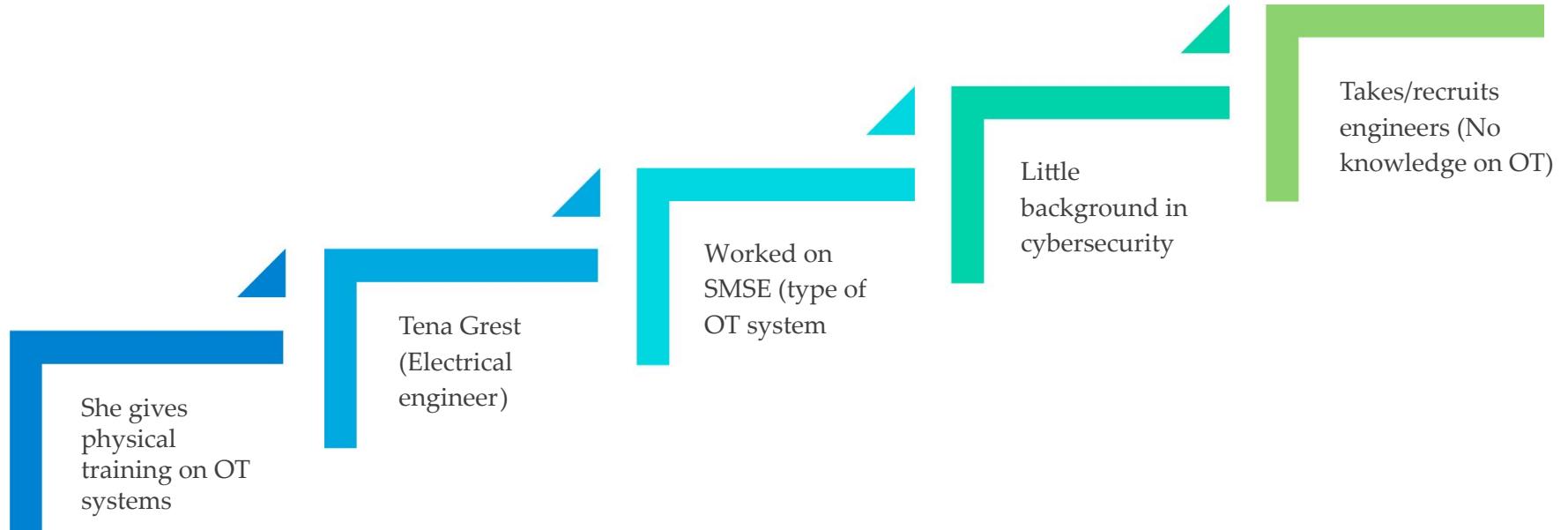
Value Proposition Canvas

BRIAN



Beneficiary Workflow 1 (Tena)

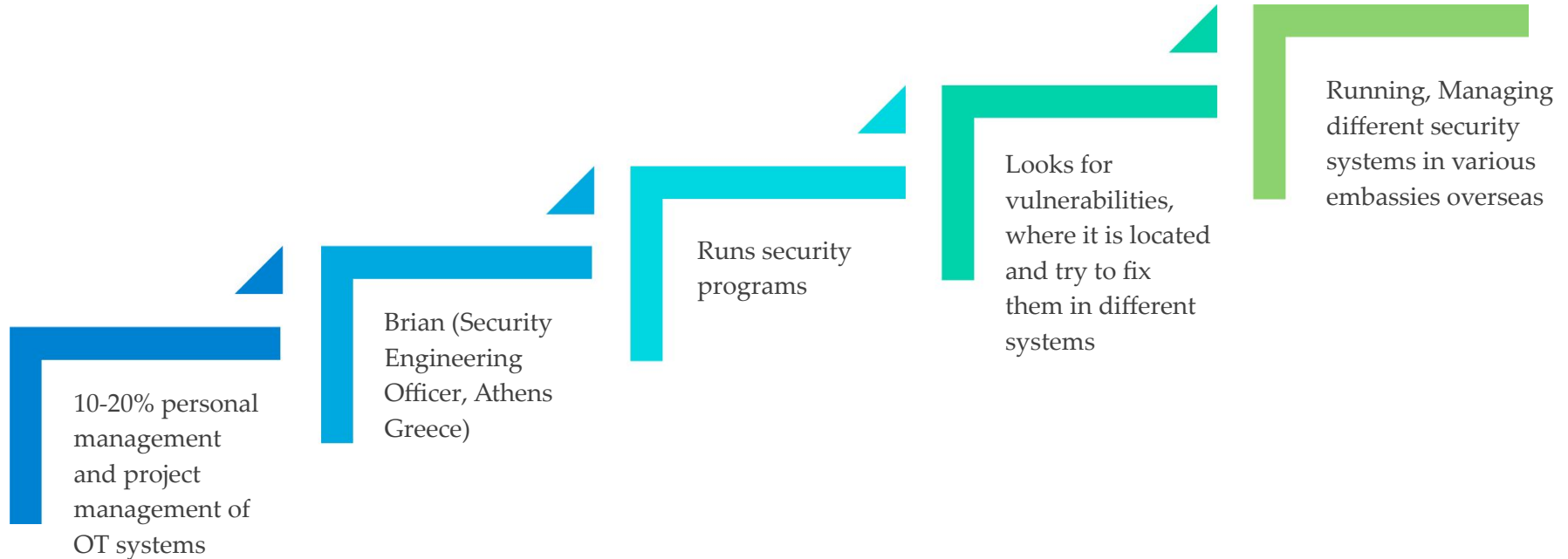
(Day to day activities)



Her end goal is she wants a “system” that helps the learning and instructing

Beneficiary Workflow 2 (Brian)

(Day to day activities)



Works more on physical security than Cybersecurity.

MVP + Experiments

Which software/tool can we put here?

Our MVP this week are 3 softwares

What is it used for?

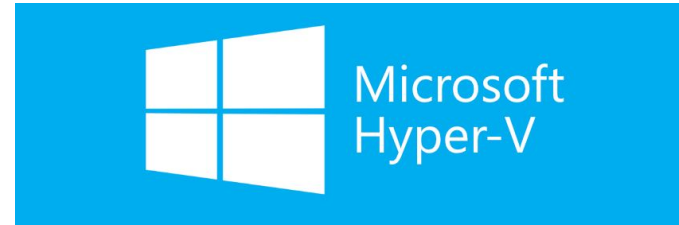
To simulate a virtual lab training environment

What did we expect?

- Can they handle multiple users at once?
- Can it connect to the internet?
- Can they provide hands-on training?

What did we get?

It might be possible to include specific features in these softwares through the help of Linux servers, web based access and access controls



Next week plan's

Plan for next week

Get to know more about the features and capabilities of the softwares

Who will you talk to?

Risk Management team, Other engineers who work on OT systems

What will you do?

Install and test out the softwares we mentioned, talk with more beneficiaries

What do you need to find out?

Is the free version of the softwares good for a demo?



Week 6

Interviews this week: **2** (Phillip), (Ben Hansen)

Total Interview counts: **9**

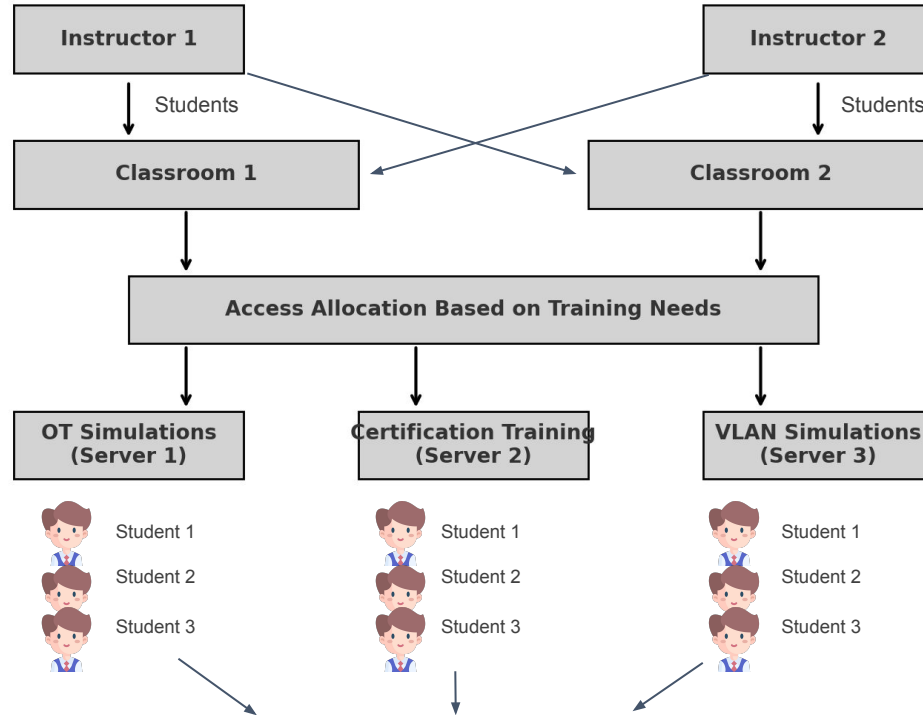
Updated Problem Statement

There is a need for a “virtual platform” to provide better training for people at the DoS.



FROM A NON-TECHNICAL POINT OF VIEW

Virtual Training Network Flow



N number of students can access these training simultaneously

Beneficiary Discovery Week 6 (What we learnt from the beneficiary interviews)



Hypotheses

What did you think beginning the week?

When it comes to hands on training, does it only involve OT systems or something else?



Experiments

What did you think beginning the week?

We interviewed Phillip and Ben Hansen to gain insights into the problem statement and daily operations of various DoS units.



Results

What did you discover?

We as a team have better understanding of the problem. A virtual training platform can deliver hands-on training on different environments.



Actions

What's next? How do you proceed?

The hardware/software requirements for our proposed solution.

Beneficiary Discovery Proof (Our discussion was virtual as they live in DC and Virginia)

[External] Re: Beneficiary Interview

This event occurred 7 hours ago (Fri 3/7/2025 3:30 PM - 4:00 PM) Chat

Microsoft Teams Meeting

Hansen, Benjamin L <HansenBL@state.gov> invited you [Accepted 1, Didn't respond 4](#)

Follow up Follow up

You accepted on 3/7/2025 3:18 PM

CC: Mosalikanti, Krishna Chaitanya <kmosalik@iu.edu>; Karunakaran, Vinoth <vkarunag@iu.edu>; Balaji, Thangamuthu <tbalaji@iu.edu>; Gerst, Tena M <Gerstm2@state.gov>

Subject: Re: [External] Re: Beneficiary Interview

You don't often get email from rajanan@iu.edu. [Learn why this is important](#)

Thank you for your email Ben,

We as a team are available at 3:30 pm on Friday. Would that be a convenient time to talk?

Thanks
Ananmay

Get [Outlook for iOS](#)

Hide message history

Hansen, Benjamin L <HansenBL@state.gov> sent a meeting request

Hansen, Benjamin L <HansenBL@state.gov>

To: Rajan, Ananmay

Cc: Mosalikanti, Krishna Chaitanya; Karunakaran, Vinoth; Balaji, Thangamuthu; +1 other

[External] DS-56

GM

Gerst, Tena M <Gerstm2@state.gov> Wed 3/5/2025 9:55 AM

To: Rajan, Ananmay; Hansen, Benjamin L <HansenBL@state.gov>

Cc: Karunakaran, Vinoth; Balaji, Thangamuthu; Mosalikanti, Krishna Chaitanya; +1 other

Some people who received this message don't often get email from gerstm2@state.gov. [Learn why this is important](#)

This message was sent from a non-IU address. Please exercise caution when clicking links or opening attachments from external sources.

Ananmay,

Please meet (virtually) Ben Hansen. He leads the Risk Mitigation Engineering team and is much more versed on the world of OT than I am. I hope he will be able to answer some of the technical questions that I could not.

Cheers,

Tena

Ms. Tena Gerst
Office Director
Office of Technical Security Engineering
Teams: 771-204-3735 Mobile: 202-256-1287
Email: gerstm2@state.gov

Beneficiary Discovery (Continued)

[External] Accepted: DS-56 Project Meeting with Phillip

This event occurred 3 days ago (Tue 3/4/2025 3:00 PM - 3:30 PM) Chat

Microsoft Teams Meeting

Accepted 1, Didn't respond 3

Idea 1.docx Idea 2.docx Idea 2.docx

Follow up

You sent a meeting request

Rajan, Ananmay

To: Mosalikanti, Krishna Chaitanya; Karunagaran, Vinoth; Balaji, Thangamuthu; +1 other Tue 3/4/2025 10:45 AM

Microsoft Teams [Need help?](#)

[Join the meeting now](#)

Meeting ID: 235 489 227 630

Passcode: Gq6Dp3B9

Join on a video conferencing device

Tenant key: teams@vc.iu.edu

Video ID: 116 725 998 8

[More info](#)

For organizers: [Meeting options](#)

The Mission Model Canvas (Week 6)



Key Partner

Phillip Miyasaka
(Problem Sponsor)
Tena Grest, Ben Hansen



Key Activities

Making a virtual platform to
integrate simulations
Completed 9 rounds of
interviews



Value Proposition

1. To audit OT systems.
2. Gain practical knowledge on various environments (OT, Networks)
3. Hands-on training to protect the organization's Network from cyber threats
4. Hands-on security training through certificates.



Buy-in & Support

WIP



Deployment

WIP



Mission Budget

They are open to investing in third party vendors, softwares, etc



Achievement

Providing a way for hands-on training in a virtual setting



Beneficiaries

1. OT security personnel
2. Engineers working in Springfield, VA
3. Technical Security Personnel
4. People with no cybersecurity background



Key Resources

- Red team/Blue Team Simulation Framework
- Establish basic network security testing environment.
- Internal Capacity Building Programs.

Value Proposition Canvas - (Phillip)

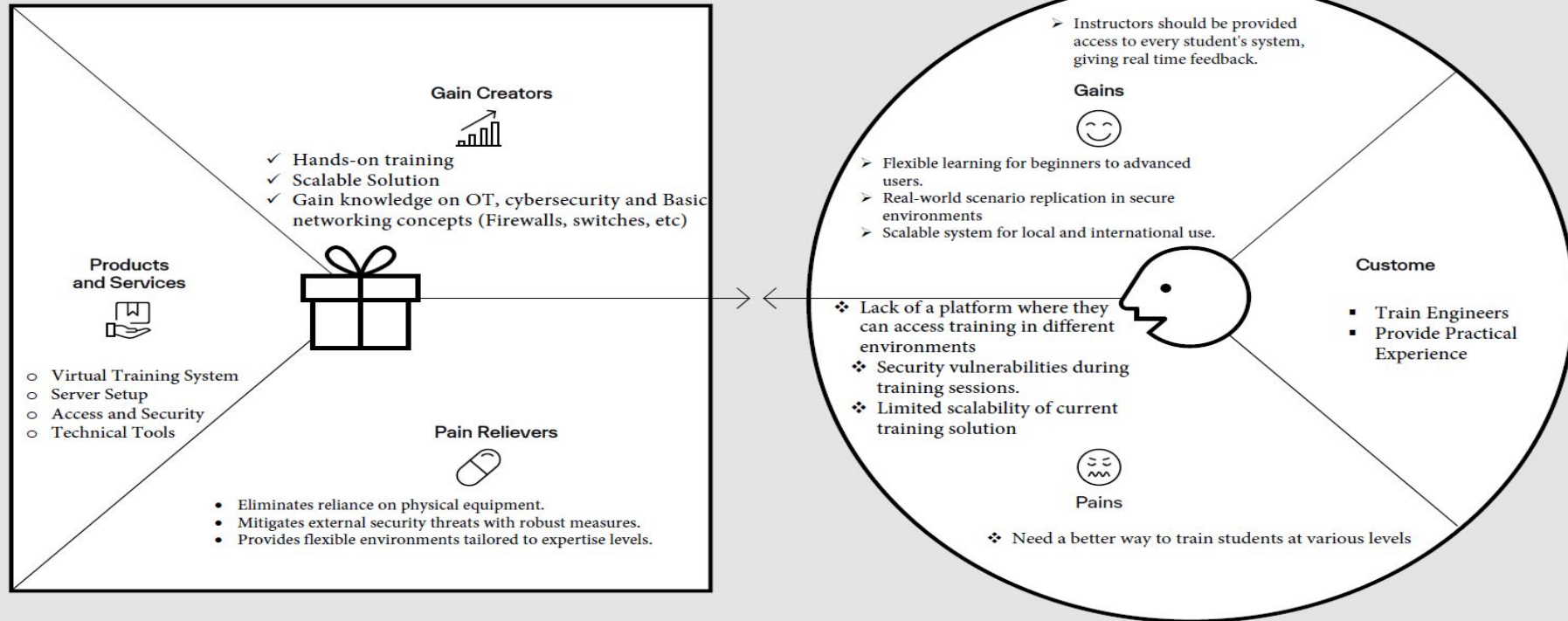
Phillip (Project Sponsor)

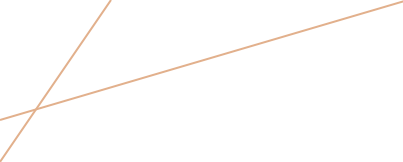
The Value Proposition Canvas

Value Proposition:

Flexible, scalable, and secure virtual training platform providing hands-on experience in cybersecurity, OT systems, and IT concepts.

Customer Segment: Engineers in Springfield, Virginia (Department of State) and potential global users (U.S. embassies) seeking practical training in OT security and IT certifications.





Beneficiary Archetype for Phillip

- Can be described as “proactive training visionary with practical concerns”.
- He is a visionary leader, problem solver, security conscious, very practical in nature, collaborative stakeholder and a result oriented leader.

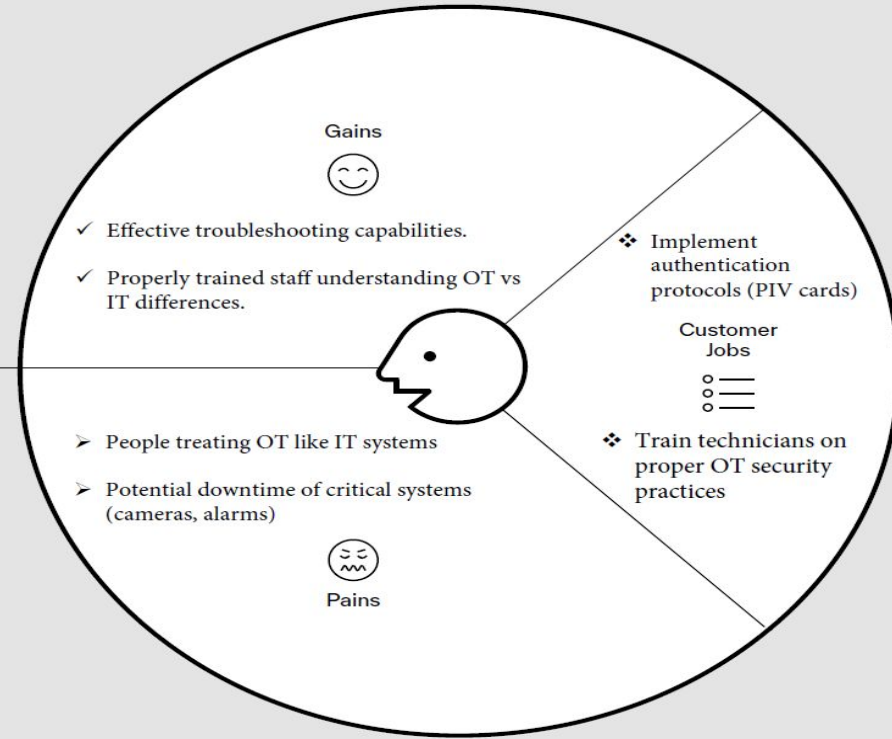
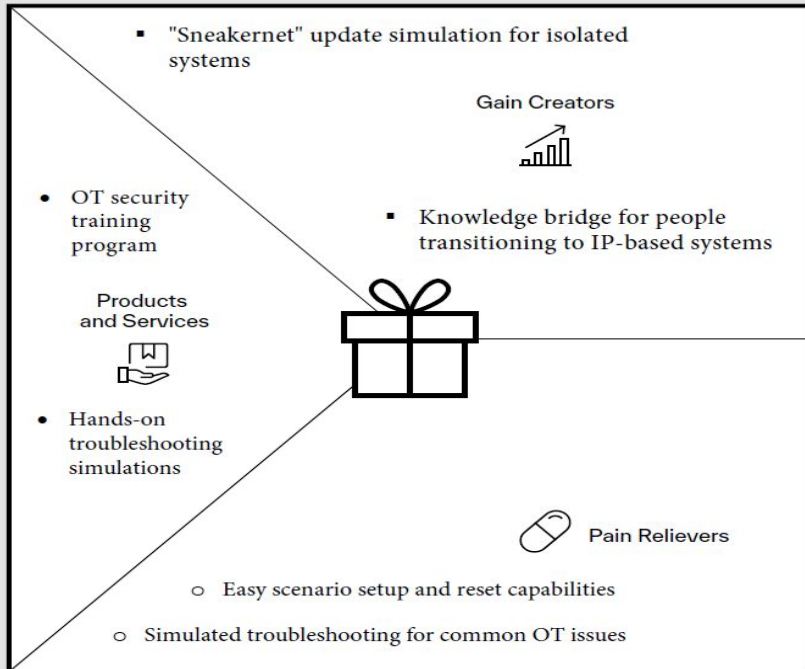
Value Proposition Canvas - (Ben Hansen)

Ben Hansen

The Value Proposition Canvas

Value Proposition: Specialized OT Security Training Program for DoS Risk Mitigation

Customer Segment: Risk Mitigation Unit and OT Security Personnel at Department of State



Beneficiary Archetype for Ben

Ben Hansen

ABOUT

Oversees operational technology security across DoS facilities, focusing on maintaining critical systems like security cameras and fire alarms. Prioritizes operational continuity over constant updates, understanding the severe consequences of OT system downtime.

DEMOGRAPHIC INFORMATION

- Location: Springfield, VA
- Role: Risk Mitigation Officer

MOODS AND PERSONALITY

(Friendly, disciplined, strong)

DAY-TO-DAY WORKFLOW

- Works primarily on operational technology security with focus on continuity over constant updates

GOALS AND NEEDS

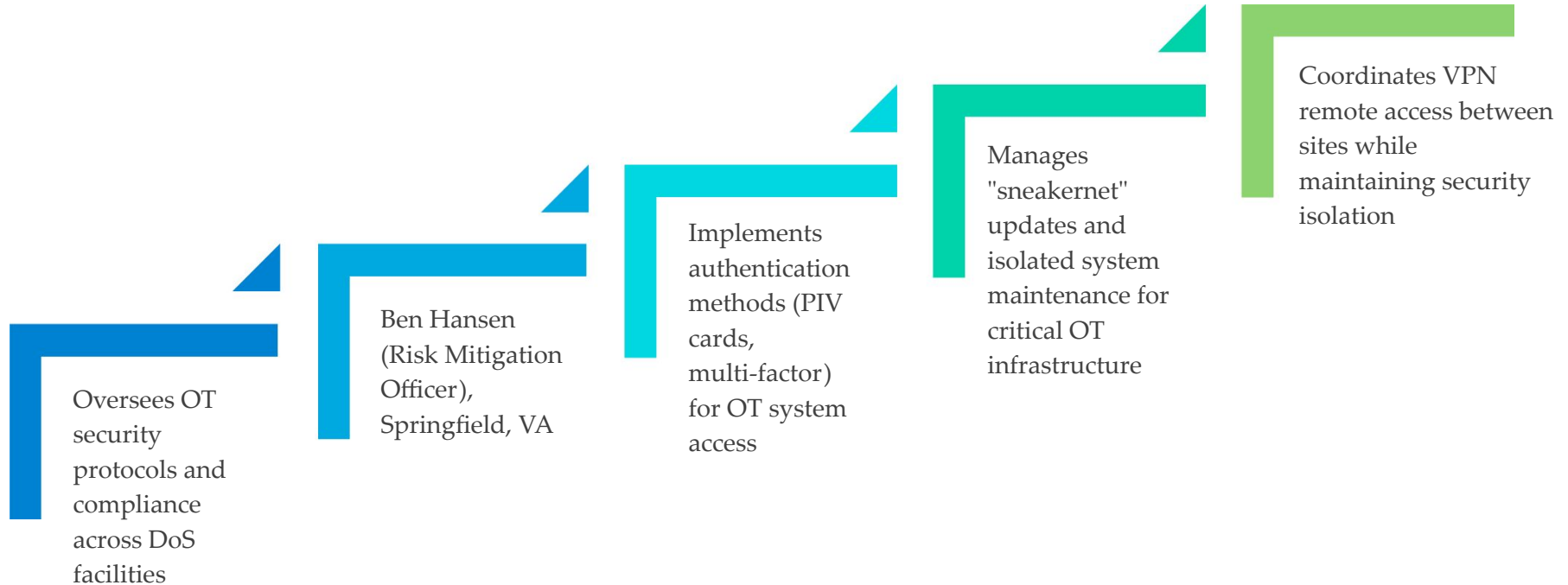
- Ensure proper separation between OT and IT systems
- Maintain continuous operation of critical security systems
- Implement secure authentication methods for OT access
- Train technicians on proper OT security practices
- Manage secure updates for isolated systems

CHALLENGES

- People treating OT systems like IT systems when they require different security approaches
- Maintaining clear separation between OT and IT security boundaries
- Preventing downtime in critical systems like security cameras and fire alarms

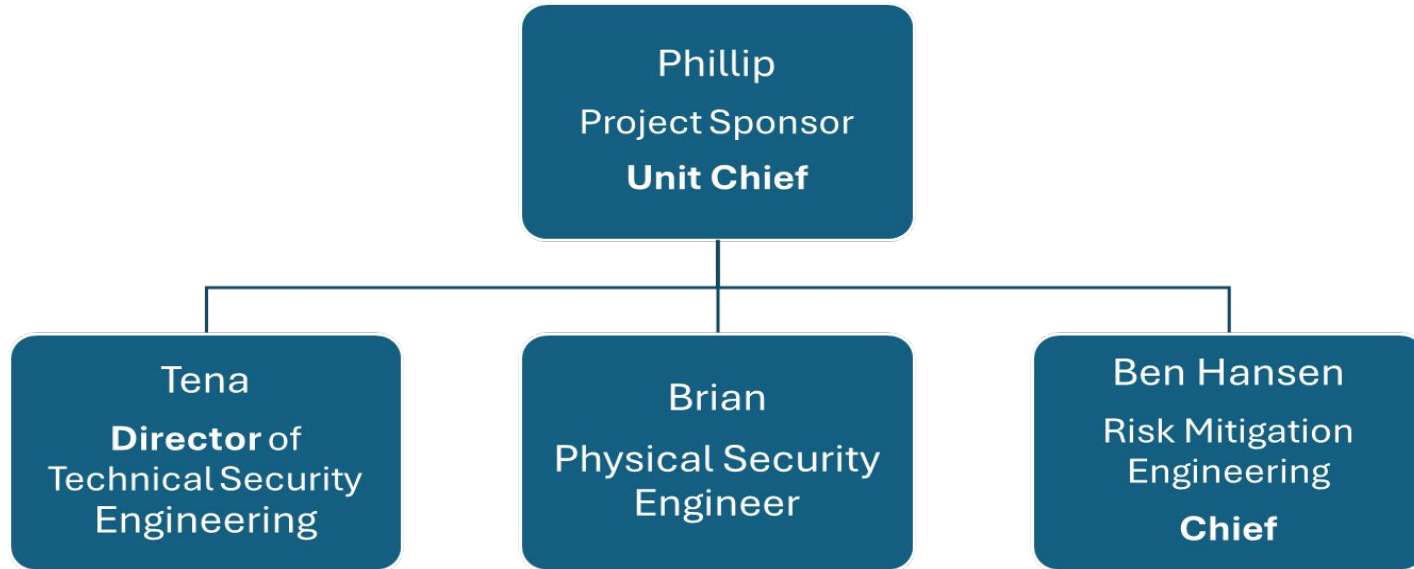
Beneficiary Workflow (Ben Hansen)

(Day to day activities)



Works primarily on operational technology security with focus on continuity over constant updates.

Organizational Chart (Draft)



MVP + Experiments Conducted (Week 6)

MVP Experiment Canvas

Project name:

Made by:

Ananmay
Rajan

Start date/time:

End date/time:

4. Customer Engagement ❤️

How are you going to engage the customers who used your MVP, and gather feedback to improve your Value Proposition?

Follow-up
email / call

1. Your Customer Segment 👤

Which segment of your target customers will experience the most value from your MVP, and can you learn from them?

2. Value Proposition 📋

My company, _____ (insert of company) is developing _____ (a defined offering) to help _____ (a defined audience) _____ (solve a problem) with _____ (secret sauce/differentiation) _____.

3. Channel(s) 🗣️

How are you going to reach your Customer Segment with your Value Proposition to them?

Virtual platform
with secure
access

5. Riskiest Assumption(s) 🎯

What you are setting out to test with your MVP?

Are these assumptions about the Problem, Solution or Implementation?

6. Experiment Format 🧪

WHAT type of experiment are you going to run?

Simulated virtual training labs managed manually.

A key goal is to provide a windows-based user interface.

Test with small groups of OT personnel, staff at various levels @DoS & system administrator based in Springfield, VA

7. Scenario / Workflow 🔄

Users access the virtual platform via secure login.

Instructors assign specific tasks in the virtual lab (e.g., OT simulations, red/blue team exercises).

Users complete tasks while monitored by instructors for feedback.

System logs performance metrics and user feedback.

8. Metrics 📊

What

Multiple User Access.

Feedback scores on the platform effectiveness.

9. Success Criteria 🏆

How do you qualify and/or quantify a successful outcome of this experiment?

At least 50-60 users complete the training with positive feedback

10. Results (Expected) 📈

V
R

Pilot training with 60 users.

Collect data on usability, effectiveness, and scalability.

Simulate Red vs Blue Team scenarios and measure the results.

11. Learnings & Insights 🧠

What are your key learnings?

What are your key surprises?

Did you get enough results?

What do they tell you about your Riskiest Assumption(s)?

Assess if virtual training meets the needs of technical and non-technical personnel.

VALIDATED

INVALIDATED

INCONCLUSIVE

12. Next steps 🚀

What do the results tell you to do next?

Are you going to Pivot, Pursue, or Stop?

Find out different possible solutions to the current problem

Next week plan's (Week 6)



Plan for next week

Identify one or more solution to the existing problem.



Who will you talk to?

We will be talking to more DoS staff who are located in Springfield, VA to get to know more about their current challenges and training requirements.



What will you do?

How can we make a training network with a limited space (7 racks)



What do you need to find out?

See if VLAN training is actually something what they want in the network

Understand the various POV's of different people at the DoS on a training network. Do they prefer it?

Week 9

Interviews this week: **3**

Total Interview counts: **14**

Updated Problem Statement

There is a need for a “virtual platform” to provide better training for students and instructors at the DoS.



BENEFICIARY DISCOVERY

HYPOTHESIS

What did you think beginning the week?

When it comes to hands on training, does it only involve OT systems or something else?

EXPERIMENTS

What did you do this week?

Reached out to multiple beneficiaries to know their thoughts on our progress.

RESULTS

What did you discover?

We as a team have better understanding of the problem. A virtual training platform can deliver hands-on training on different environments.

ACTIONS

What's next? How do you proceed?

The hardware/software requirements for our proposed solution.

Pivot or Proceed?

We are going with Proceed.

We continued to validate the hypothesis through beneficiary discovery. The interviews and feedback confirmed that there is a strong need for a virtual training platform for both students and instructors at the Department of State.

Next Step

- Continue interviewing key stakeholders to gather insights on hardware/software preferences and limitations.
- Develop a prototype virtual platform with features identified during the discovery phase.
- Test the platform to validate its usability and effectiveness for both **technical** and **non-technical training**.
- Engage with more beneficiaries to validate the prototype and incorporate the feedbacks.



Beneficiary Discovery Proof (Our discussion was virtual as they live in DC and Virginia)

[External] RE: Our update regarding the project DS-56

Sorry for the delay, I've been teaching lately so haven't had time on my email.

Concept looks good as long as the hardware and network can support the load.

To answer your questions.

1. Who are the main instructors who do the teaching? What do they feel they lack in terms of training or knowledge?
Diplomatic Security Certified Trainers will be teaching and be the main instructors as well as Management personnel.
2. Do you feel simulations would be a good way of teaching? If the simulations can mimic the real environment with full control and access to vender specific software then yes.
3. When it comes to training, we understand that right now students are going through software's and courses available online, is there something else that is being done as well or is that it? Currently, students are being taught via online, inperson, and vendor training. We want a platform that can do real situational training on vender specific software for OT systems. In-person commercial certifications will be taught on this system as well.
4. Does every student need to undergo all sorts of training that is included in our "virtual system" or is it only specific to specific people?

Below are the difference classes we would teach on this system.

Operational Technology Systems

Operational Technology Security Officer course

CompTIA Network+

CompTIA Tech +

CompTIA A+

CompTIA Security+

CompTIA CYSA

CompTIA Security X

ISACCA CISA

Maybe CISSP

Hope this answers your questions and uses.

Thanks,

Questions related to DS-56 project



Mosalikanti, Krishna Chaitanya

To: HansenBL@state.gov

Cc: ☺ Rajan, Ananmay



Sat 3/29/2025 10:45 PM

Hi Ben, as you know, we are group DS-56 working on a "virtual platform" to better train students and help instructors. We would like to ask you a couple of questions based from our research and it would be helpful if you could provide some insights on them.

Q1. Since your work involves mostly OT systems within the DoS, what type of training do you feel is lacking for students? We thought of going ahead with a simulation-based virtual training to get that "hands-on feel" to train on different OT systems. Do you feel that is something that would benefit in terms of teaching and learning and is something that hasn't been implemented before?

Q2. Could you provide us with a day-to-day timeline of your work, what do you do and at any point during your work do you feel our project can make a difference?

Q3. Do the students (Different engineers based on our interviews with other DoS staff) lack the "security" training aspect when working on OT systems?

Q4. In your opinion, do you think using third party services for simulations and software's to train students is good for the DoS? How much control is exactly needed in that case?

Q5. Any suggestions would you like to give to help our group in successfully building a "virtual platform" (which will be built using 7 racks as mentioned by Philip) especially in terms of OT training?

Let us know your thoughts on these and have a good day!

Regards,
Krishna Chaitanya

Beneficiary Discovery (Continued)

Questions related to DS-56 project

 Mosalikanti, Krishna Chaitanya

To: gersttm2@state.gov

Cc: 🌐 Rajan, Ananmay

Sat 3/29/2025 10:45 PM

Hi Tena, as you know, we are group DS-56 working on a virtual platform to better train students and help instructors. We would like to ask you a couple of questions based from our research and it would be helpful if you could provide some insights on them.

Q1. Do you mainly teach security aspects of different systems present in the DoS or does it involve something else as well?

Q2. Do you feel security training on different environments (i.e. OT systems, VLAN's, etc.) is missing in your current training curriculum? If so, do you believe a simulation-based training is a good way to teach?

Q3. Could you provide us day-to-day timeline of your work, and at which point you are facing "difficulties" or feel something better can be done related to this project?

Q4. Did the "students" face any problems with the current teaching methods when they try to study and are they facing any issues trying to study complex topics? It would be great to know their thoughts on this "virtual platform" we are trying to create and different concepts they are studying.

Q5. Are there any personal suggestions for us to improve our "virtual platform" from a teacher point of view as well as students?

Let us know your thoughts on these and have a good day!

Regards,
Krishna Chaitanya

Questions related to DS-56 project

 Mosalikanti, Krishna Chaitanya

To: Hayes8C@state.gov

Cc: 🌐 Rajan, Ananmay

Sat 3/29/2025 10:45 PM

Hi Brian, as you know, we are group DS-56 working with Philip Miyasaka on a "virtual platform" to better train students and help instructors. We would like to ask you a couple of questions based from our research and it would be helpful if you could provide some insights on them.

Q1. Since you have been a security engineering officer for approximately 13 years, we would like to know what all cybersecurity concepts can be incorporated into the "virtual platform" we are trying to build to better safeguard DoS at Springfield, VA against any external cyber-attacks. Our virtual platform includes VM's, Hypervisor's and using third party software's/simulations as of right now.

Q2. Could you provide us with a day-to-day timeline of your work, what do you do and at any point during your work do you feel or would like to have a "virtual platform" which would benefit your unit to get trained on different environments? Right now, we are focusing on training on OT environments, VLAN simulations and using the internet to access certificate-based training like CISCO's, etc.

Q3. At your workplace, are there any curriculums being used right now to teach people about the environments that I mentioned? If there aren't any, do you just give them the necessary training for your assigned unit? Do they only cover the security side of things?

Q4. Are there any personal suggestions for us to improve our "virtual platform" from a security perspective?

Let us know your thoughts on these and have a good day!

Regards,
Krishna Chaitanya

THE MISSION MODEL CANVAS (MMC) WEEK 9

Key Partner

Phillip Miyasaka
(Problem Sponsor)
Tena Grest (Instructor), Ben
Hansen (Our OT guy)

Key Activities

- Making a virtual platform to integrate simulations.
- Keeping it secure.

Value Proposition

1. To audit OT systems.
2. Sandbox style training on various environments. (OT, Networks)
3. Get real-time monitoring and feedback.
4. Hands-on security training through certificates.

Buy-in & Support

WIP

Deployment

WIP

Beneficiaries

1. OT security personnel (students)
2. Engineers working in Springfield, VA (students)
3. Instructors at the DoS
4. People with no cybersecurity background (Students)

Key Resources

- Red team/Blue Team Simulation Framework
- Establish basic network security testing environment.
- Internal Capacity Building Programs.

Mission Budget

They are open to investing in third party vendors, softwares, etc.

Achievement

Providing a way for hands-on trainings in a virtual setting.

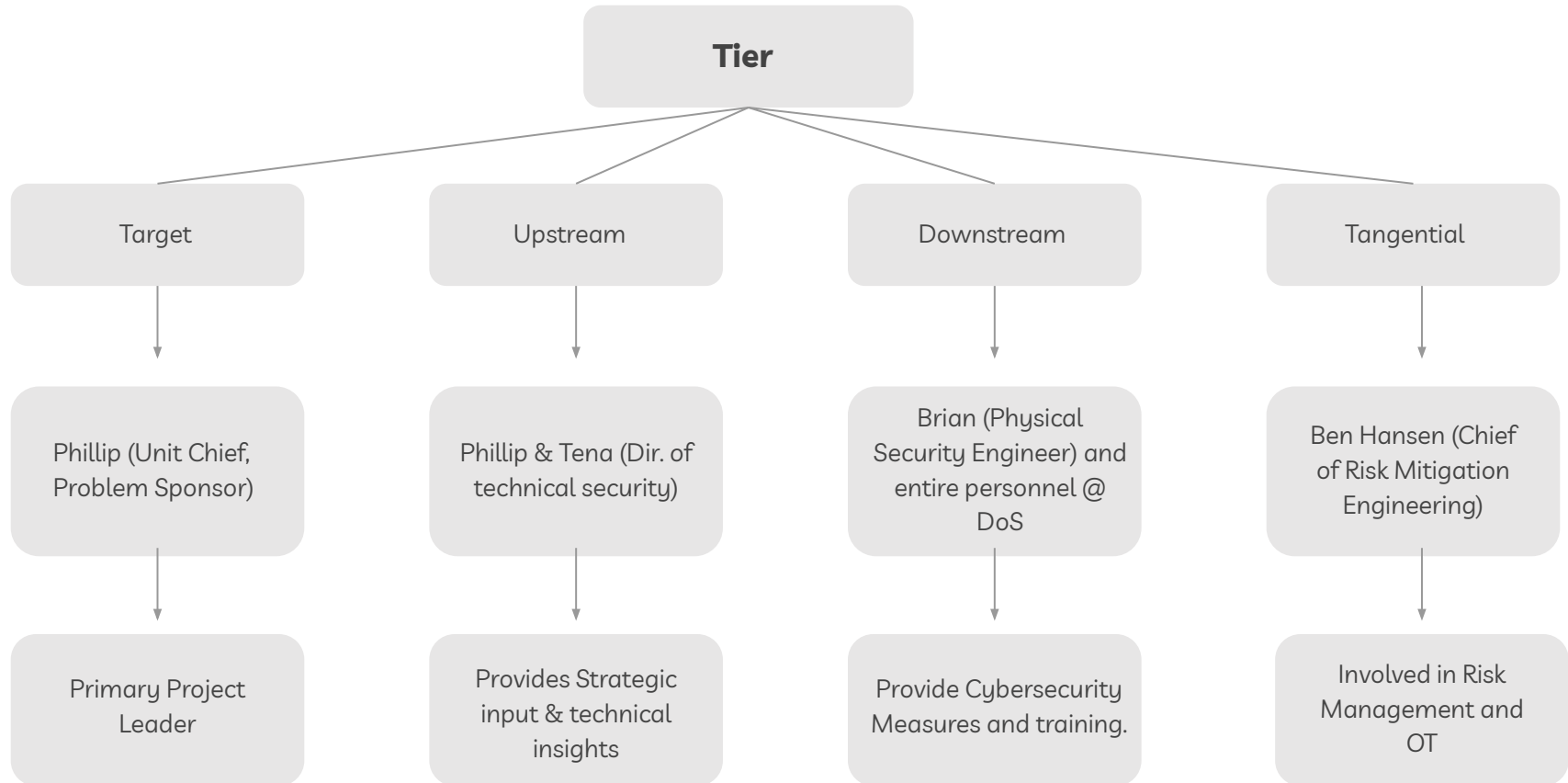
Beneficiary Evolution

Philip

1. Initially, we had no idea about the training they did (lot of ambiguity).
2. Post discussion, we amended our problem statement based on the feedback from beneficiaries.

- 
3. Further discussion and in-depth analysis revealed that this extends beyond just the OT platform, encompassing various other environments including Commercial and Network systems.
 4. Covers every embassy - Specific to DoS (Springfield, VA)

Tiered Beneficiary Diagram



VALUE PROPOSITION CANVAS - (PHILIP, INSTRUCTOR)

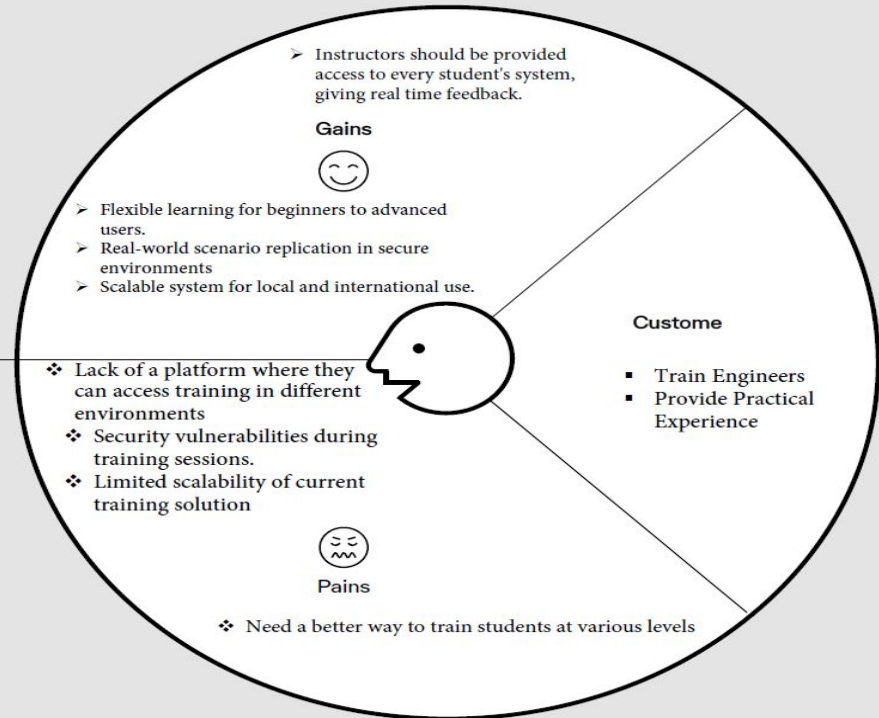
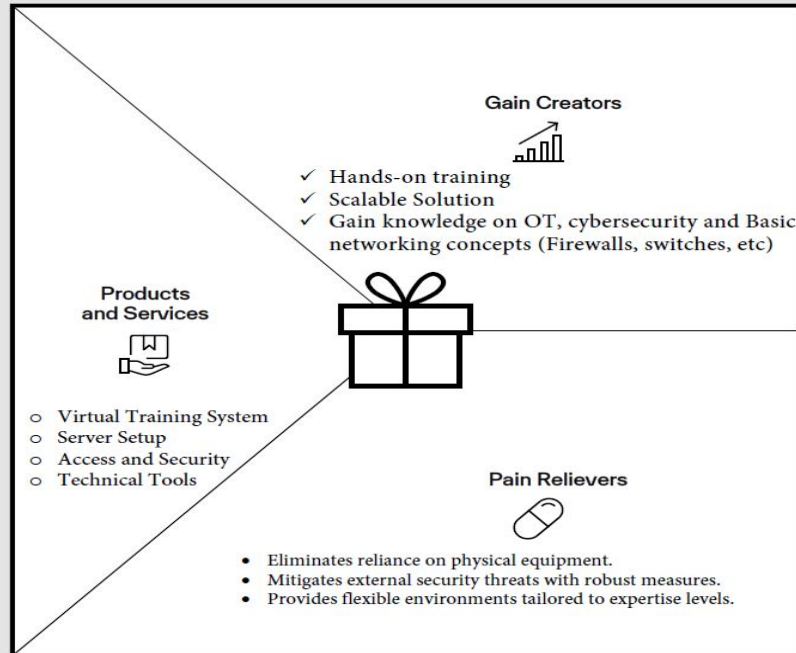
Phillip (Project Sponsor)

The Value Proposition Canvas

Value Proposition:

Flexible, scalable, and secure virtual training platform providing hands-on experience in cybersecurity, OT systems, and IT concepts.

Customer Segment: Engineers in Springfield, Virginia (Department of State) and potential global users (U.S. embassies) seeking practical training in OT security and IT certifications.



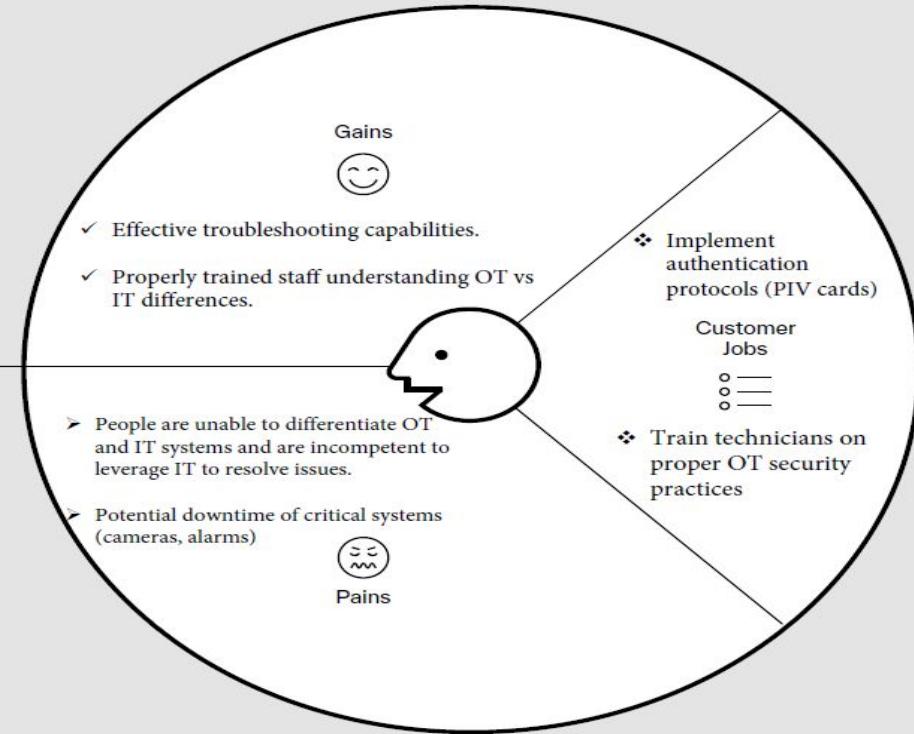
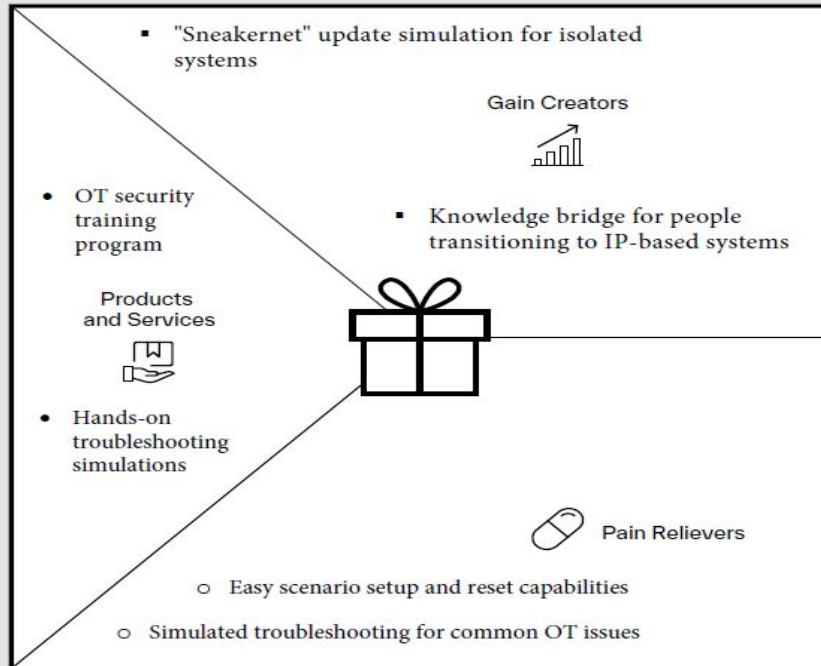
VALUE PROPOSITION CANVAS - (BEN HANSEN)

Ben Hansen

The Value Proposition Canvas

Value Proposition: Specialized OT Security Training Program for DoS Risk Mitigation

Customer Segment: Risk Mitigation Unit and OT Security Personnel at Department of State



Beneficiary Archetype for Ben

Ben Hansen

ABOUT

Oversees operational technology security across DoS facilities, focusing on maintaining critical systems like security cameras and fire alarms. Prioritizes operational continuity over constant updates, understanding the severe consequences of OT system downtime.

DEMOGRAPHIC INFORMATION

- Location: Springfield, VA
- Role: Risk Mitigation Officer

MOODS AND PERSONALITY

(Friendly, disciplined, strong)

DAY-TO-DAY WORKFLOW

- Works primarily on operational technology security with focus on continuity over constant updates

GOALS AND NEEDS

- Ensure proper separation between OT and IT systems
- Maintain continuous operation of critical security systems
- Implement secure authentication methods for OT access
- Train technicians on proper OT security practices
- Manage secure updates for isolated systems

CHALLENGES

- People treating OT systems like IT systems when they require different security approaches
- Maintaining clear separation between OT and IT security boundaries
- Preventing downtime in critical systems like security cameras and fire alarms

WEEK 9 MVP + EXPERIMENTS

MVP Experiment Canvas

Project name:

Advancing Operational
Technology Training
for Diplomatic
Missions

Made by:

Ananmay
Rajan

4. Customer Engagement ❤️

How are you going to engage the customers who used your MVP, to learn from their experience and gather feedback to improve your Value Proposition?

Follow-up
email / call

1. Your Customer Segment 👤

Which segment of your target customers will experience the most value from your MVP, and can you learn from them?

- OT security personnel
- Engineers working in Springfield, VA
- Technical Security Personnel
- People with no cybersecurity background

2. Value Proposition 📄

My company, _____ (insert of company) _____ is developing _____ (a defined offering) _____ to help _____ (a defined audience) _____ (solve a problem) _____ with _____ (secret sauce/differentiation) _____.

A virtual training environment at DoS that bridges the technical skills gap, offering standardized training in OT, Commercial, and VLAN with real-time instruction and performance tracking on a secure platform.

3. Channel(s) 🗣️

How are you going to reach your Customer Segment and "deliver" your Value Proposition to them?

Virtual platform
with secure
access

5. Riskiest Assumption(s) 🚨

What you are setting out to test with your MVP?

Are these assumptions about the Problem, Solution or Implementation?

Potential data leak
when expanding to U.S
Embassies Worldwide

6. Experiment Format 🧪

WHAT type of experiment are you going to run?

- Simulated virtual training labs managed manually.
- Goal: provide a Windows-based user interface.
- Test with small groups of OT personnel and other staff @DoS in Springfield, VA.

7. Scenario / Workflow 🔄

HOW does that work? Describe the steps from beginning to end.

- Users access the virtual platform via secure login.
- Instructors assign specific tasks in the virtual lab (e.g., OT simulations, red/blue team exercises).
- Users complete tasks while monitored by instructors for feedback.
- System logs performance metrics and user feedback.

8. Metrics 📊

What will you measure? (Can be multiple metrics)

- Multiple User Access.
- Feedback scores on how well the platform works.

9. Success Criteria 🏆

How do you qualify and/or quantify a successful outcome of this experiment?

At least 60 users complete
the training with positive
feedback

10. Results (Expected) 📈

What are the qualitative and / or quantitative

- Train 20 users, then expand to 60.
- Gather usability and effectiveness data.
- Simulate Red vs Blue Team scenarios and measure outcomes.

11. Learnings & Insights 🗨️

What are your key learnings?

What are your key surprises?

Did you get enough results?

What do they tell you about your Riskiest Assumption(s)?

Assess if virtual training
meets the needs of technical
and non-technical personnel.

VALIDATED

INVALIDATED

INCONCLUSIVE

12. Next steps 📌

What do the results tell you to do next?

Are you going to Pivot, Pursue, or Stop?

Find out different
possible solutions to
the current problem

Next week plan's (Week 9)



Plan for next week

Hardware/Software requirements for our virtual platform.



Who will you talk to?

Speak to people who may have a background with VM's, Hypervisor, physical servers to build this platform.



What will you do?

Start with the creation of our virtual platform. (Demo).



What do you need to find out?

How much it will cost for all hardwares.

Week 10

Beneficiaries Interview

Interviews this week: 3

Total Interview counts: **25**

Updated Problem Statement

There is a need for a "virtual platform" to provide better training for students and instructors at the DoS.



Beneficiary Discovery

HYPOTHESIS

What did you think beginning the week?

Is the hardware we identified (physical servers, virtual machines, and hypervisors) both cost-effective and practical for implementation in DoS?

EXPERIMENTS

What did you do this week?

We made a document for all the hardware requirements needed for our "virtual platform" and our sponsor went through it and gave a few suggestions.

RESULTS

What did you discover?

The hardware components for physical servers, VM's and Hypervisors are affordable, scalable and robust for now.

ACTIONS

What's next? How do you proceed?

We need to find the softwares/simulations to give that hands-on training for different environments.

The Mission Model Canvas

Key Partner

Phillip Miyasaka
(Problem Sponsor)
Tena Grest (Instructor),
Ben Hansen (Our OT guy)

Key Activities

- Making a virtual platform to integrate simulations.
- Keeping it secure.

Value Proposition

1. To audit OT systems.
2. Sandbox style training on various environments.
3. (OT, Networks)
4. Get real-time monitoring and feedback.
5. Hands-on security training through certificates.

Buy-in & Support

WIP

Deployment

WIP

Beneficiaries

1. OT security personnel (students)
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Key Resources

- Red team/Blue Team Simulation Framework
- Establish basic network security testing environment.
- Internal Capacity Building Programs.

Mission Budget

They are open to investing in third party vendors, softwares, etc.

Achievement

Providing a way for hands-on training in a virtual setting.

Dual-Use Summary

Private Sector

- Manufacturing Industries (Energy, Manufacturing, Utilities, Critical infrastructure)
- Healthcare

Team Analysis

- For manufacturing industries they might need OT and cybersecurity training for automating and controlling production lines, assembly processes, and robotic operations.
- For healthcare, they might need OT based training in laboratories and healthcare equipment. Also, commercial training is essential to undergo HIPAA training and how to secure patient and other health related data under EHR (Electronic Health Record).

Conclusion

Manufacturing

- Pros: Our project is well-suited for the manufacturing sector, as these industries require OT training to secure and automate production lines, robotic operations, and critical systems ensuring minimal disruption
- Cons: Include the complexity of integrating with legacy systems.

Healthcare

- Pros: Our virtual platform's Operational Technology (OT) and cybersecurity training provides significant benefits to healthcare organizations. This training enhances their ability to protect sensitive patient health information and strengthens their defenses against potential cyber threats. By implementing these security measures, healthcare providers can better safeguard critical data while reducing vulnerability to future cyber attacks.
- Cons: Adapting to new technologies. (Managing user access to our training platform: A comprehensive guide.).

Mission Achievement



Phillip (Instructor @ DoS)

A prototype virtual platform for DoS students and instructors that provides hands-on training with vendor-specific operational technology systems. The platform enables realistic situational training and supports in-person commercial certification programs.



Ben Hansen (Risk Mitigation Officer @ DoS)

An educational environment that effectively bridges the gap between operational technology and information technology systems, creating a unified training environment.



Brian (Physical Security Engineer @ DoS, Athens, Greece)

A virtual platform that trains users to perform operational activities correctly and consistently before starting their role, with high-fidelity simulation of DoS operational technology systems.



Tena (Instructor @DoS)

Can access and view student's progress during training to give real-time feedback and monitoring.

Mission Achievement Metrics

For Philip, we measure success of the training platform through **certification pass rates and demonstrated skill proficiency.**

For Ben Hansen, we track progress in **bridging the OT/IT gap through knowledge improvement and collaborative problem-solving.**

For Brian, we assess the **realism and effectiveness of OT simulation training through system fidelity and error reduction.**

For Tena, we evaluate **instructor capabilities through monitoring coverage and feedback effectiveness.**

VALUE PROPOSITION CANVAS - (PHILIP, INSTRUCTOR)

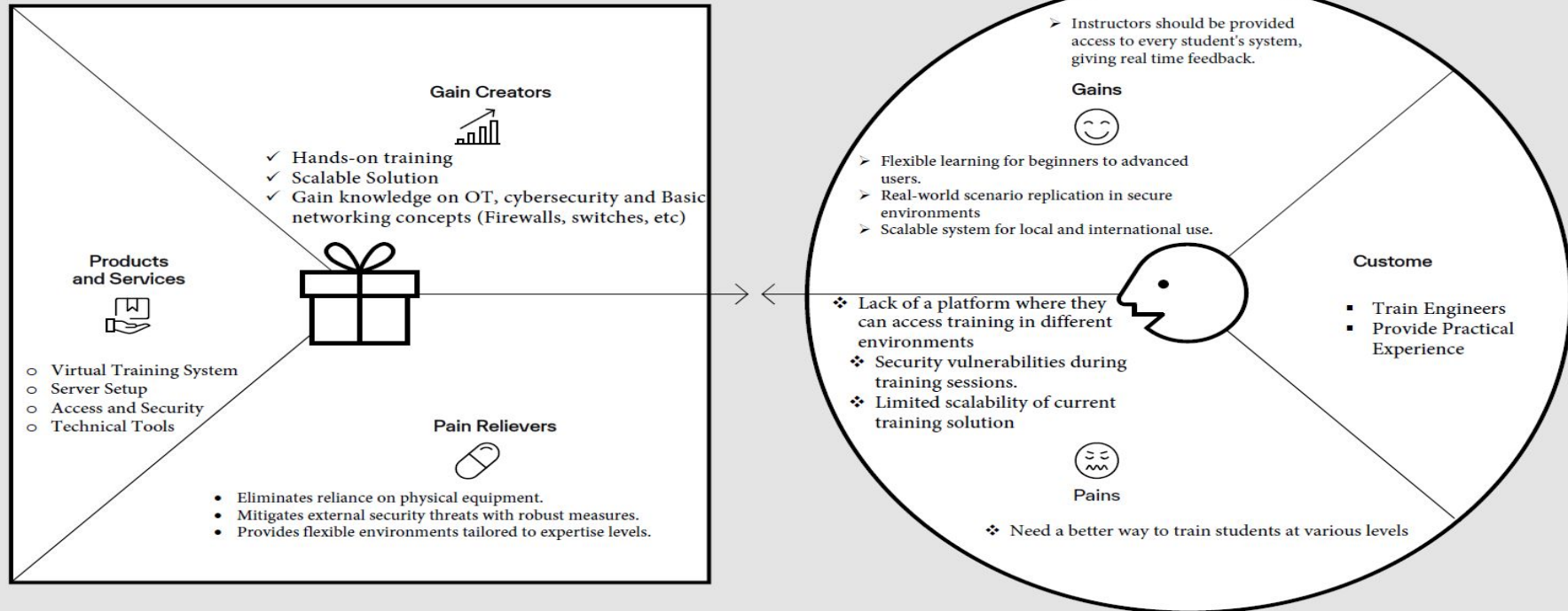
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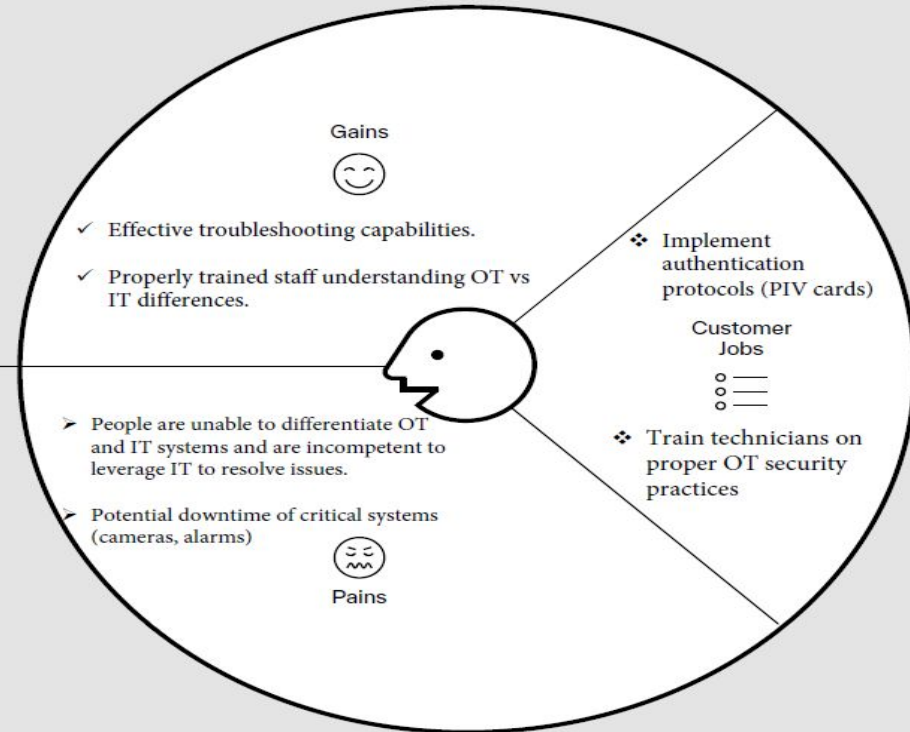
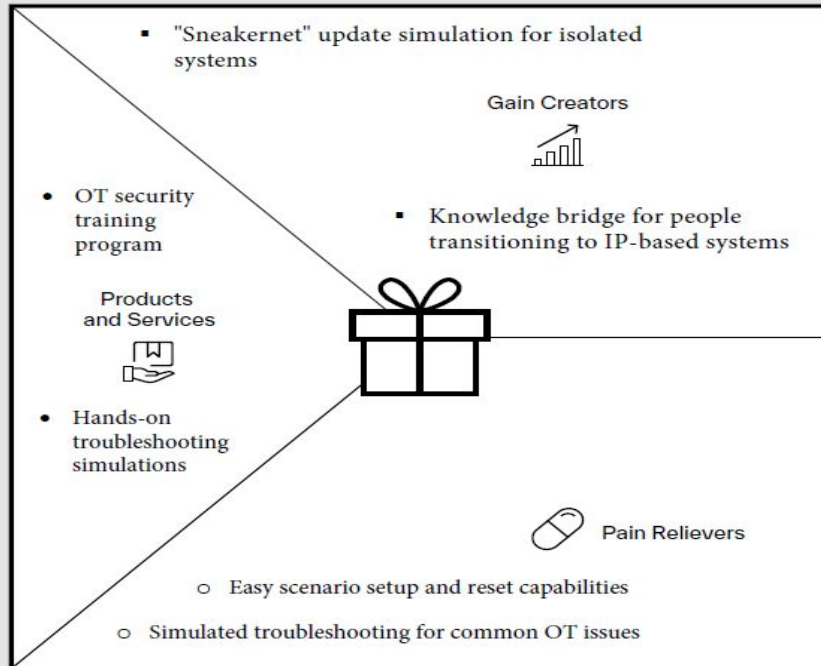
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WEEK 10 MVP + EXPERIMENTS CONDUCTED

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Missions

Made by:

Ananmay
Rajan

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Follow-up
email / call

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VALIDATED

INVALIDATED

INCONCLUSIVE

12. Next steps 🚀

What do the results tell you to do next?
Are you going to Pivot, Pursue, or Stop?

Find out different possible solutions to the current problem

MVP (Contd..) (Hardware Requirements for now)

128 vCPUs, 768 GB RAM, and
NVMe SSD storage



20-25 VM's

Our hypervisor - Proxmox VE

128 vCPUs, 768 GB RAM, and
NVMe SSD storage



20-25 VM's

Our hypervisor - Proxmox VE

(Physical Servers for OT simulations)

Use Case

- ✓ Simulating real-world cyber-attacks and training on OT systems.
- ✓ Running multiple simulations simultaneously.
- ✓ Instructor has access to multiple VM's through Hypervisor.

For each VM

CPU: 4 vCPUs (For running multiple simulation processes)

RAM: 8-12 GB (for advanced security simulations)

Storage: 100-150 GB SSD/NVMe (if storing logs or running databases)

Network: 1 Gbps or 10 Gbps virtual NIC (for high-speed data exchange)



Virtual Machines

Next week plan's (Week 10)

Plan for next week

Software/Simulations requirements for our virtual platform.

Who will you talk to?

Speak to people who may have a background with simulations, VM's, Hypervisor, physical servers to build this platform.

What will you do?

Start with the creation of our virtual platform. (Demo).
Make a "cost analysis document" for all hardware and software requirements.

What do you need to find out?

How much it will cost for all softwares?

Week 11

Beneficiaries Interview

Interviews this week: 3

Total Interview counts: 25

Updated Problem Statement

There is a need for a "virtual platform" to provide better training for students and instructors at the DoS.



Beneficiary Discovery

HYPOTHESIS

What did you think beginning the week?

Why do they want VLAN training?

EXPERIMENTS

What did you do this week?

Had a meeting with our Project Sponsor (Phillip) to understand more on the OT trainings and its security aspect as well as understanding VLAN training in depth.

RESULTS

What did you discover?

VLAN training will involve a separate secure network system that mimics the DoS current security network. (i.e. Honeypots for Security purpose).

OT training (Optional), main goal is to provide training on Physical OT devices.

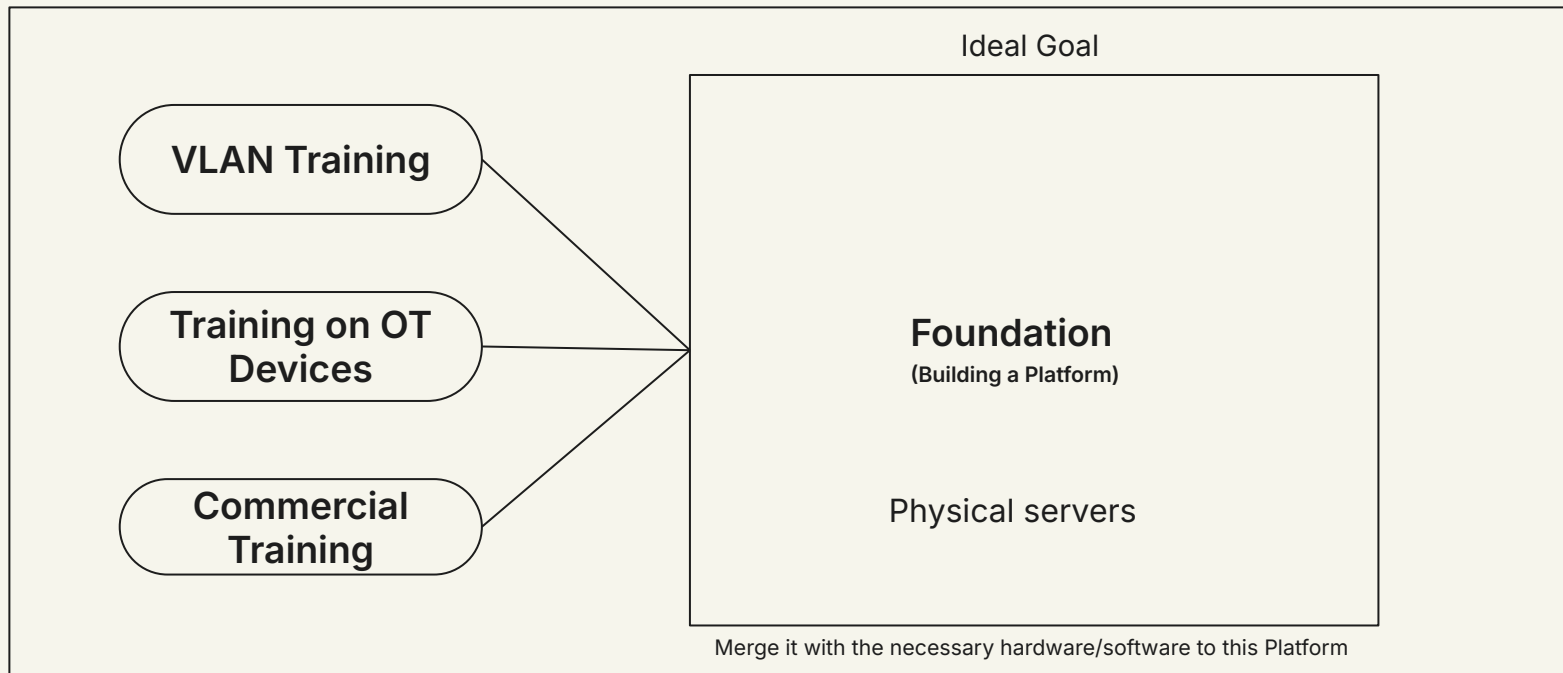
ACTIONS

What's next? How do you proceed?

We need to find the softwares to give that hands-on training for Building Management systems (BMS) for OT training.

Ideation Mapping

"Virtual"



The Mission Model Canvas

Key Partner

Phillip Miyasaka
(Problem Sponsor)
Tena Grest (Instructor),
Ben Hansen (Our OT guy)

Key Activities

- Making a virtual platform to integrate different environments.
- Keeping it secure.

Value Proposition

1. To audit OT systems.
2. Sandbox style training on various environments.
3. (OT, Networks)
4. Get real-time monitoring and feedback.
5. Hands-on security training through certificates.

Beneficiaries

1. OT security personnel (students)
2. Engineers working in Springfield, VA (students)
3. Instructors at the DoS
4. People with no cybersecurity background (Students)

Key Resources

- Technical Infrastructure (like Physical servers and VMs)
- Software Resources (like 3rd vendor softwares for OT environments)
- Knowledge Resources (Content and materials)

Buy-in & Support

- Leadership Support
- Financial & Technical Resources
- Stakeholder & Compliance Elements.

Deployment

- Implementation Phases.
- Technical Infrastructure
- User Adoption Strategy

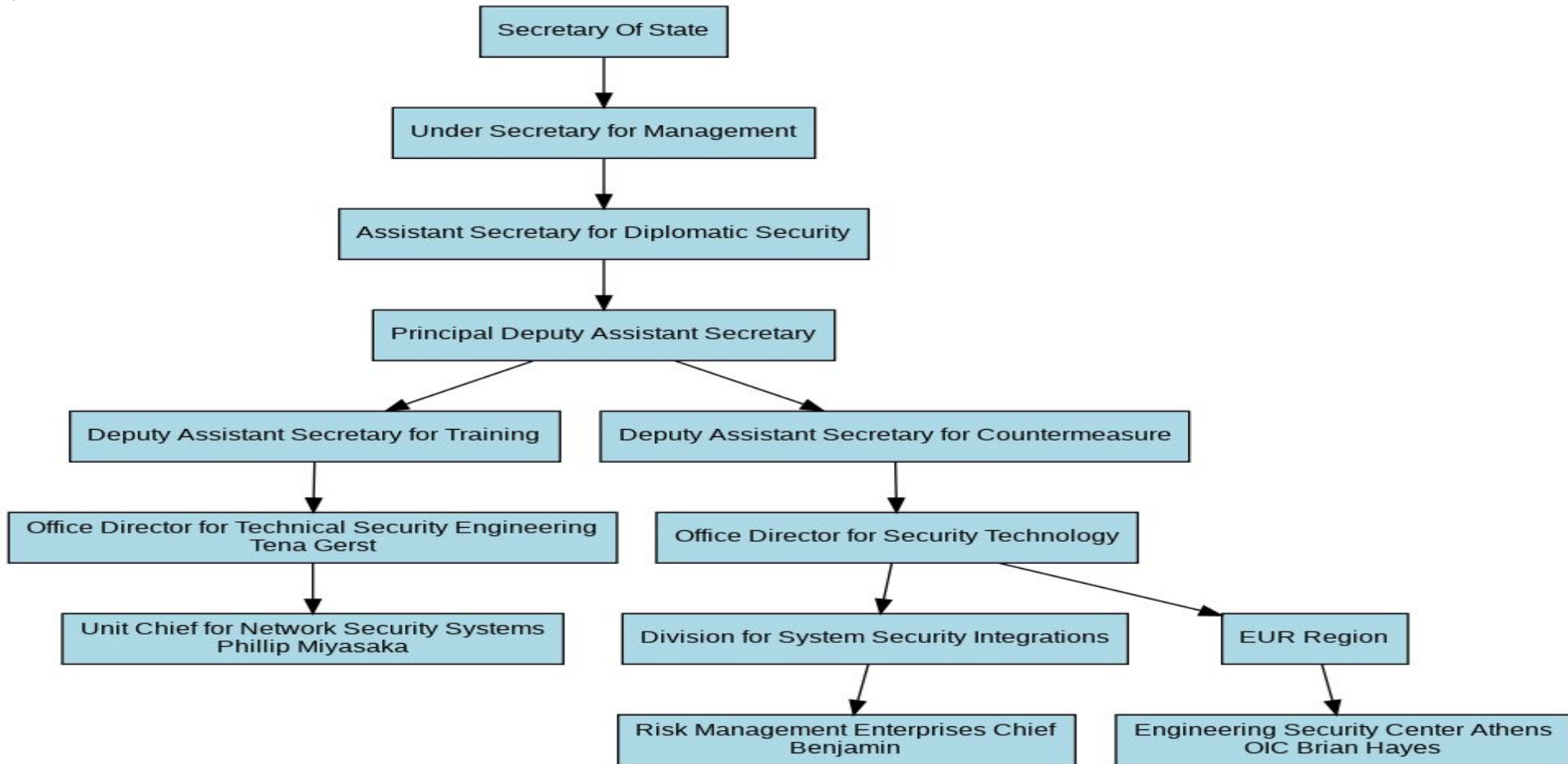
Mission Budget

\$500,000 allocated for building the entire Virtual Network System.

Achievement

Providing a way for hands-on training in a virtual setting.

Organizational Chart/Influence Chart



Buy-in / Support story for your product

Explain who are the necessary people to rapidly deploy your product and how you are going to win them over.

- Different embassies
- DoS Personnel in Springfield, VA for initial testing and validation
- Technical insights from external tech organizations
- IT Security Managers and Training Program Managers to facilitate internal support and implementation.

How are you going to win them over?

- A training aid specific to activities/work in DoS.
- Provide a targeted demonstration emphasizing real-world, hands-on simulations, tailored explicitly to DoS operational workflows.

Get-Keep-Grow Diagram



Get

- U.S Embassies worldwide connected to DoS
- Educational universities (i.e. RIT & CMU)



Keep

- Smooth user experience
- Maintain reliable system Performance



Grow

- Expand with cloud-based labs and services (future)
- Scale to international embassies beyond U.S

VALUE PROPOSITION CANVAS - (PHILIP, INSTRUCTOR)

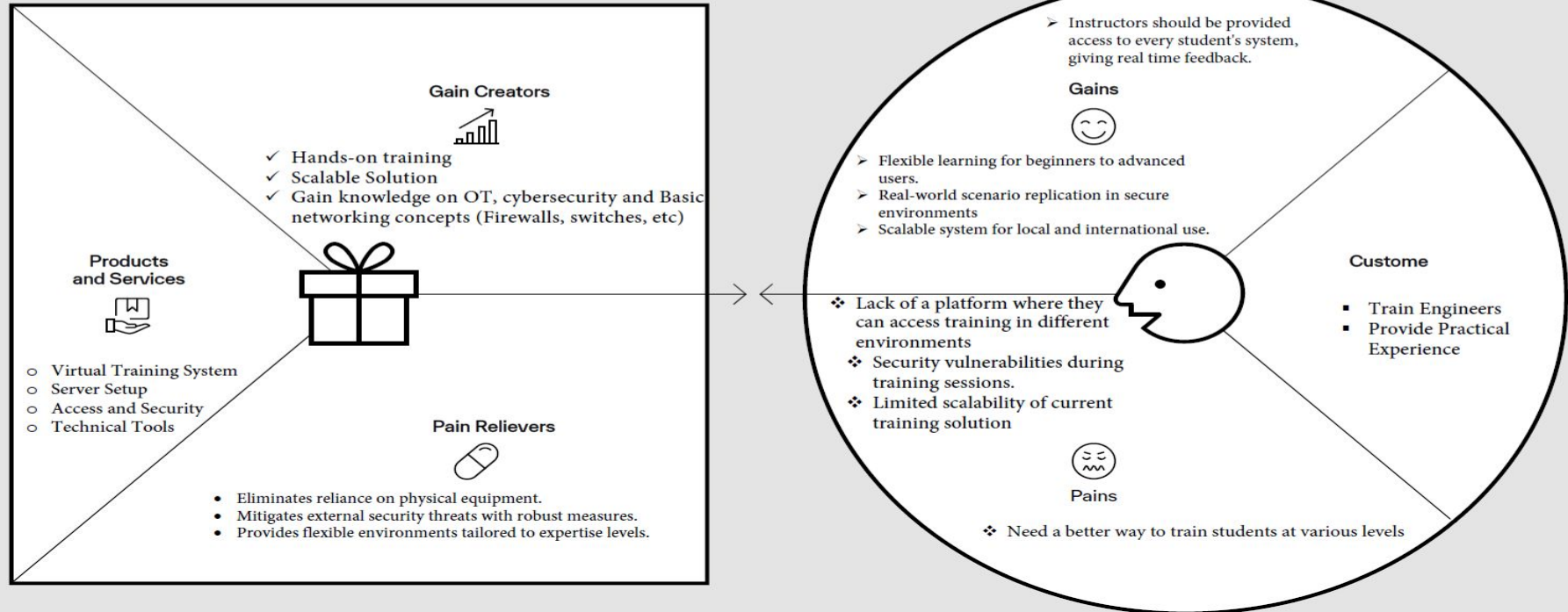
Phillip (Project Sponsor)

The Value Proposition Canvas

Value Proposition:

Flexible, scalable, and secure virtual training platform providing hands-on experience in cybersecurity, OT systems, and IT concepts.

Customer Segment: Engineers in Springfield, Virginia (Department of State) and potential global users (U.S. embassies) seeking practical training in OT security and IT certifications.



Beneficiary Archetype for Phillip

Phillip Miyasaka

ABOUT THE USER

Phillip is a senior engineer overseeing the design and implementation of OT security training at the Department of State. He works with both technical and non-technical teams and is focused on modernizing training infrastructure across global U.S. embassies.

DEMOGRAPHIC INFORMATION

- Location: Springfield, VA
- Occupation: Unit Chief - Network Security Systems

MOODS AND PERSONALITY

(Friendly, disciplined, strong)

DAY-TO-DAY WORKFLOW

- Coordinates with instructors and engineers on OT security practices
- Reviews training methods, system vulnerabilities, and compliance tools
- Evaluates new technologies for training and security improvement

CHALLENGES

- No unified platform for hands-on OT and cybersecurity training
- Security risks from decentralized or outdated training methods
- Limited scalability and adaptability in current training systems

GOALS AND NEEDS

- Deliver secure, scalable, and flexible training for a wide range of DoS personnel
- Replace physical equipment with virtual labs to reduce costs and risks
- Monitor training sessions in real-time to ensure quality and consistency

POTENTIAL BUY-IN

- **Customization:** Tailor content to DoS workflows and allow instructor control over simulations
- **Metrics That Matter:** Show how the platform can track certification progress, compliance readiness, and reduce training gaps
- **Scalability Vision:** Present a roadmap showing how the solution can grow across international embassies

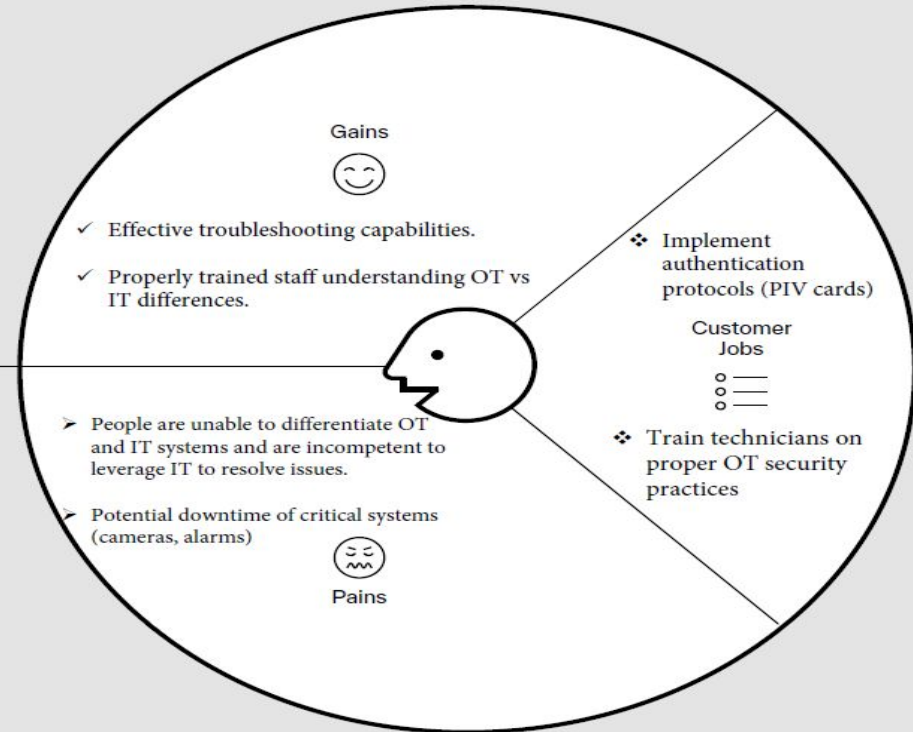
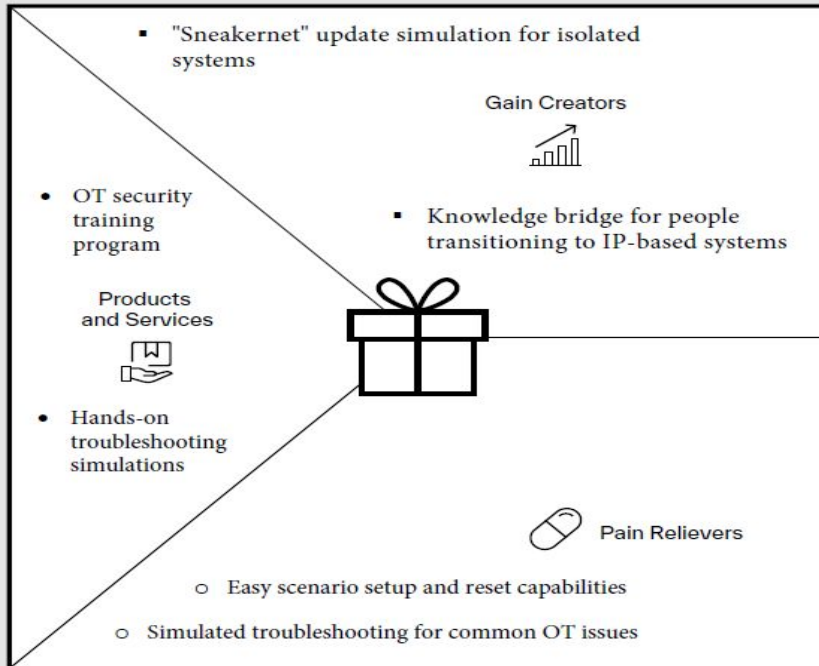
VALUE PROPOSITION CANVAS - (BEN HANSEN)

Ben Hansen

The Value Proposition Canvas

Value Proposition: Specialized OT Security Training Program for DoS Risk Mitigation

Customer Segment: Risk Mitigation Unit and OT Security Personnel at Department of State



Beneficiary Archetype for Ben

Ben Hansen

ABOUT

Oversees operational technology security across DoS facilities, focusing on maintaining critical systems like security cameras and fire alarms. Prioritizes operational continuity over constant updates, understanding the severe consequences of OT system downtime.

DEMOGRAPHIC INFORMATION

- Location: Springfield, VA
- Role: Risk Mitigation Officer

MOODS AND PERSONALITY

(Friendly, disciplined, strong)

DAY-TO-DAY WORKFLOW

- Works primarily on operational technology security with focus on continuity over constant updates

GOALS AND NEEDS

- Ensure proper separation between OT and IT systems
- Maintain continuous operation of critical security systems
- Implement secure authentication methods for OT access
- Train technicians on proper OT security practices
- Manage secure updates for isolated systems

CHALLENGES

- People treating OT systems like IT systems when they require different security approaches
- Maintaining clear separation between OT and IT security boundaries
- Preventing downtime in critical systems like security cameras and fire alarms

WEEK 11 MVP + EXPERIMENTS CONDUCTED

MVP Experiment Canvas

Project name:

A virtual platform delivering hands-on training across different environments.

Made by:

Ananmay Rajan

4. Customer Engagement ❤️

How are you going to engage the customers who used your MVP, to learn from their experience and gather feedback to improve your Value Proposition?

Follow-up email / call

1. Your Customer Segment 👤

Which segment of your target customers will experience the most value from your MVP, and can you learn from them?

2. Value Proposition 📄

My company, __ (insert of company) __, is developing __ (a defined offering) __ to help __ (a defined audience) __ (solve a problem) __ with __ (secret sauce/differentiation) __.

3. Channel(s) 🗣️

How are you going to reach your Customer Segment and "deliver" your Value Proposition to them?

Virtual platform with secure access

5. Riskiest Assumption(s) 🎯

What you are setting out to test with your MVP?

Are these assumptions about the Problem, Solution or Implementation?

Insufficient data sanitization measures resulting in residual data exposure between training sessions.

6. Experiment Format 🧪

WHAT type of experiment are you going to run?

- Simulated virtual training labs managed manually.
- Goal: provide a Windows-based user interface.
- Test with small groups of OT personnel and other staff @DoS in Springfield, VA.

7. Scenario / Workflow 🔄

HOW does that work? Describe the steps from beginning to end.

Secure Login: Users access the platform securely.

Task Assignment: Instructors assign tasks (e.g., simulations, team exercises).

Real-Time Monitoring: Instructors monitor activities and provide feedback.

Performance Logging: System logs user performance and feedback.

Temporary VM Storage: Servers store VM data temporarily for review, then erase securely.

8. Metrics 📊

What will you measure? (Can be multiple metrics)

- Multiple User Access.
- Feedback scores on how well the platform works.

9. Success Criteria 🏆

How do you qualify and/or quantify a successful outcome of this experiment?

At least 60 users complete the training with positive feedback

10. Results (Expected) 📈

What are the qualitative and / or quantitative

- Train 20 users, then expand to 60.
- Gather usability and effectiveness data.
- Simulate Red vs Blue Team scenarios and measure outcomes.

11. Learnings & Insights 🧠

What are your key learnings?

What are your key surprises?

Did you get enough results?

What do they tell you about your Riskiest Assumption(s)?

Assess if virtual training meets the needs of technical and non-technical personnel.

VALIDATED

INVALIDATED

INCONCLUSIVE

12. Next steps 🚀

What do the results tell you to do next?

Are you going to Pivot, Pursue, or Stop?

See if the platform is scalable, flexible and robust for simultaneous use

MVP (Contd..) (Hardware Requirements for now)

128 vCPUs, 768 GB RAM, and
NVMe SSD storage



20-25 VM's

Our hypervisor - Proxmox VE

128 vCPUs, 768 GB RAM, and
NVMe SSD storage



20-25 VM's

Our hypervisor - Proxmox VE

(Physical Servers for OT training)

Use Case

- ✓ Simulating real-world cyber-attacks and training on OT systems.
- ✓ Running multiple simulations simultaneously.
- ✓ Instructor has access to multiple VM's through Hypervisor.

For each VM

CPU: 4 vCPUs (For running multiple simulation processes)

RAM: 8-12 GB (for advanced security simulations)

Storage: 100-150 GB SSD/NVMe (if storing logs or running databases)

Network: 1 Gbps or 10 Gbps virtual NIC (for high-speed data exchange)



Virtual Machines

Next week plan's (Week 11)

Plan for next week

Software requirements to give the training part.

Who will you talk to?

Speak to people who may have a background with Networks, VM's, Hypervisor, physical servers to build this platform.

What will you do?

Possibly build a prototype to show our sponsor.

Make a "cost analysis document" for all hardware and software requirements.

What do you need to find out?

Are different softwares for OT training compatible with our virtual platform?